DATA COMMUNICATION & NETWORKING

A Complete Easy Guide

For Assistant Programmer & IT Officer
Competitive Examinations in Bangladesh

BPSC • Bangladesh Bank • Combined Bank SO(IT) • BCC
NTRCA (ICT) • BREB • RAKUB • Ministry of ICT

Every concept explained in plain language.
Every formula worked out step by step.

What This Book Covers

All theory topics and every numerical type from the *Data Communication & Computer Networking* chapter of the standard written-exam guide — rewritten from scratch in simple, plain English, with fully worked mathematics, memory tricks, comparison tables and a practice question bank.

How to Use This Book

1. Read a chapter once for understanding — do not memorise on the first pass.
2. Re-read only the **Key Point** and **Easy Way to Remember** boxes. These carry the highest marks per minute.
3. Work every **Worked Example** with pen and paper. Numericals are where most candidates lose marks.
4. Two days before the exam, read only Chapter 15 (Formula Sheet & Rapid Revision) and attempt Chapter 16 (MCQ Practice).

Syllabus Map

This is how the topics in this book map to what actually gets asked. Weightage is based on the recurring pattern of past questions from BPSC AP, Combined Bank SO(IT), BCC, BREB and NTRCA (ICT) papers.

Ch	Topic	Weightage	Most Likely Question Type
1	Data Communication Fundamentals	Very High	Definitions, duplex modes
2	Signals, Bandwidth & Impairments	High	Nyquist / Shannon numerical
3	Digital & Analog Transmission	Medium	Line coding, PCM steps
4	Multiplexing & Transmission Media	High	FDM vs TDM, Cat5 vs Cat6
5	Switching & Topologies	Very High	Circuit vs packet switching
6	OSI & TCP/IP Models	Very High	7 layers with functions
7	Error Detection & Flow Control	High	CRC / Hamming numerical
8	Multiple Access, Ethernet, Devices	High	CSMA/CD vs CSMA/CA
9	**IP Addressing & Subnetting**	**Highest**	**Subnetting numerical**
10	Routing	High	Static vs dynamic, DV vs LS
11	Transport Layer	Medium	TCP vs UDP, handshake
12	Application Layer Protocols	Medium	DNS, DHCP functions
13	Network Security	Medium	Symmetric vs asymmetric
14	Wireless & Mobile Networks	Medium	802.11 standards, 2G–5G
15	Formula Sheet & Rapid Revision	—	Final revision
16–17	MCQ & Written Question Bank	—	Practice

Where Candidates Lose Marks

Three areas account for most lost marks in this subject: **(1)** subnetting numericals done without writing the binary, **(2)** confusing bit rate with baud rate, and **(3)** mixing up which OSI layer a device belongs to. Each of these has a dedicated section in this book.

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CHAPTER 1

Data Communication Fundamentals

1.1 Data, Information and Data Communication

Data is raw, unprocessed facts. The number `85`, the word `Dhaka`, a photograph — on their own these mean nothing.

Information is processed data — data that has been given meaning. "85 marks in Mathematics" is information.

Data communication is the exchange of data between two devices through some transmission medium such as a copper wire, a fibre-optic cable or radio waves.

Four Characteristics of a Good Data Communication System

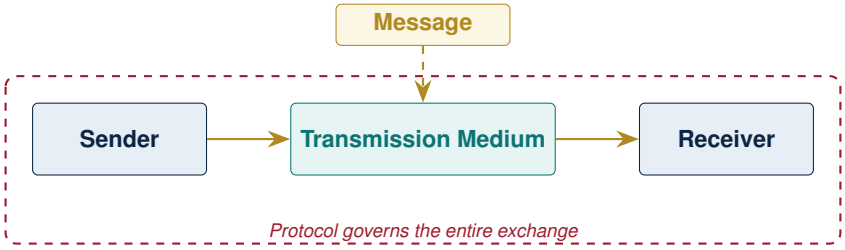
Remember these four. They are asked as a direct one-mark definition question.

- **Delivery** — data must reach the correct destination, and *only* that destination.
- **Accuracy** — data must arrive unaltered. Corrupted data is unusable.
- **Timeliness** — data must arrive on time. Late video or voice data is useless (this is called *real-time transmission*).
- **Jitter** — the variation in packet arrival time. If packets arrive at 20 ms, then 40 ms, then 25 ms, the audio or video will sound or look uneven.

Easy Way to Remember

D-A-T-J — **D**elivery, **A**ccuracy, **T**imeliness, **J**itter. Think of it as "**DATa Jitter**".

1.2 Five Components of Data Communication



1. **Message** — the information to be communicated: text, numbers, pictures, audio, video.
2. **Sender** — the device that sends the message: computer, workstation, telephone handset, video camera.
3. **Receiver** — the device that receives the message: computer, workstation, telephone, television.
4. **Transmission medium** — the physical path: twisted pair, coaxial cable, fibre-optic cable, radio waves.
5. **Protocol** — a set of agreed rules governing the communication. Without a shared protocol, two connected devices still cannot understand each other.

Easy Way to Remember

Think of posting a letter. **Message** = the letter, **Sender** = you, **Receiver** = your friend, **Medium** = the postal service, **Protocol** = the rule that the address goes on the front and the stamp on the top right. Both of you must know that rule, or the letter never arrives.

1.3 Data Flow: Simplex, Half-Duplex, Full-Duplex

Asked In

Differentiate between half-duplex and full-duplex transmission. — BPSC AP 2023
What is data communication? Define simplex, half-duplex and full-duplex. — ICT Ministry AP 2017, BKKB AP 2017
How many modes are used in data transfer through networks? Explain briefly. — Combined Bank SO(IT) 2024

1.3.1 Simplex

Communication is **one-directional only**. One device only sends; the other only receives. The entire capacity of the channel is used by the sending device.

Examples: keyboard to computer, computer to monitor, radio broadcast, television broadcast.

1.3.2 Half-Duplex

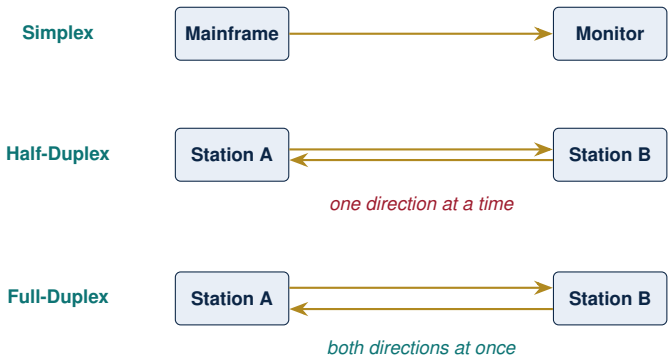
Both devices can send and receive, but **not at the same time**. When one is sending, the other must wait. The whole channel capacity is taken over by whichever device is currently transmitting.

Examples: walkie-talkie, CB radio, traditional Ethernet with a hub.

1.3.3 Full-Duplex

Both devices can send and receive **simultaneously**. The channel capacity is shared between the two directions — either by using two separate physical paths, or by dividing the bandwidth in two.

Examples: telephone conversation, mobile phone call, modern Ethernet with a switch.



Point	Simplex	Half-Duplex	Full-Duplex
Direction	One way only	Both ways, one at a time	Both ways simultaneously
Performance	—	Slower	Fastest
Channel use	Full capacity, one way	Full capacity, one device	Capacity split two ways
Example	Keyboard, TV broadcast	Walkie-talkie	Telephone
Collision	Not possible	Possible	Not possible

Easy Way to Remember

Road analogy. Simplex = a one-way street. Half-duplex = a single-lane bridge where cars from one side must wait. Full-duplex = a two-lane road with traffic flowing both ways at once.

1.4 Serial and Parallel Transmission

1.4.1 Parallel Transmission

Multiple bits (usually 8) are sent **at the same time** over n separate wires.

Advantage: fast — n times the speed of serial for the same clock rate.
Disadvantages: expensive (needs n wires), and only practical over short distances because of *skew* (bits arriving at slightly different times) and crosstalk.

1.4.2 Serial Transmission

Bits are sent **one after another** over a single wire.

Advantage: cheap, works over long distances.
Disadvantage: slower per clock cycle — but modern serial links run at such high clock rates that they now beat parallel in practice (USB, SATA, PCIe are all serial).

Common Mistake

Many candidates write "parallel is always faster". This was true historically but is *wrong today*. Parallel suffers from clock skew at high speeds, which is exactly why USB and SATA replaced the old parallel port and IDE cable. Write both the theory and this modern qualification for full marks.

1.5 Synchronous, Asynchronous and Isochronous Transmission

Asked In

Distinguish between synchronous and asynchronous transmission. — RAKUB ANSE 2023

1.5.1 Asynchronous Transmission

Data is sent **one character (byte) at a time**. Each byte is wrapped with a **start bit** (0) at the beginning and one or two **stop bits** (1) at the end. Between bytes there may be a gap of any length (idle time).

There is no shared clock. The receiver re-synchronises at every start bit.

Overhead: for 8 data bits you send at least 10 bits — a 20% overhead.

1.5.2 Synchronous Transmission

Data is sent as a **continuous stream of bits** with no start bits, no stop bits and no gaps. The bits are grouped into frames. Sender and receiver share timing, either through a common clock line or by extracting the clock from the data itself (self-synchronising line codes such as Manchester encoding).

1.5.3 Isochronous Transmission

Guarantees that data arrives at a **fixed rate with fixed delay** — essential for real-time audio and video, where an uneven arrival rate (jitter) ruins playback.

Point	Asynchronous	Synchronous
Unit of transfer	One character (byte)	Block / frame of bits
Start & stop bits	Required	Not required
Gap between units	Allowed, variable	No gap
Timing	No common clock	Common clock or self-clocking
Speed	Slower	Faster
Overhead	High (about 20%)	Low
Cost	Cheap	More expensive
Example	Keyboard, old modems	LAN, fibre links, SONET

Easy Way to Remember

Asynchronous = sending SMS messages one by one, whenever you feel like it. **Synchronous** = a marching band — everyone steps together to one beat, continuously.

CHAPTER 2

Signals, Bandwidth and Transmission Impairments

2.1 Analog and Digital Signals

Analog signal — a continuously varying wave. It has infinitely many values in its range. Human voice, temperature and the old telephone line are analog.

Digital signal — discrete steps, usually just two levels representing 0 and 1. It looks like a square wave.



Analog signal (continuous)

Digital signal (discrete)

2.1.1 Periodic and Aperiodic Signals

A **periodic** signal repeats the same pattern over a fixed interval called the **period** (T). An **aperiodic** (non-periodic) signal has no repeating pattern.

$$\text{Frequency } f = \frac{1}{T} \qquad \text{and} \qquad \text{Period } T = \frac{1}{f}$$

Frequency is measured in hertz (Hz) = cycles per second. Period is measured in seconds.

Example 2.1 — Period and Frequency

A signal has a period of 100 milliseconds. Find its frequency in kilohertz.

Step 1 — Convert to base units.

$$T = 100 \text{ ms} = 100 \times 10^{-3} \text{ s} = 10^{-1} \text{ s}$$

Step 2 — Apply the formula.

$$f = \frac{1}{T} = \frac{1}{10^{-1}} = 10 \text{ Hz}$$

Step 3 — Convert to kHz.

$$f = 10 \text{ Hz} = 10 \times 10^{-3} \text{ kHz} = \mathbf{0.01 \text{ kHz}}$$

2.1.2 Three Characteristics of a Sine Wave

- **Amplitude** — the height of the wave; its strength or power (volts, amps, watts).
- **Frequency** — how many cycles occur per second (Hz).
- **Phase** — the position of the wave relative to time zero, measured in degrees.

Easy Way to Remember

Amplitude = *how loud*. Frequency = *how high-pitched*. Phase = *where the wave starts*. All three modulation families (ASK, FSK, PSK) come from changing exactly one of these three.

2.2 Bandwidth, Bit Rate, Baud Rate and Throughput

2.2.1 Bandwidth

Bandwidth has *two different meanings* and the exam tests whether you know both.

Bandwidth in hertz — the range of frequencies a channel can carry. If a line passes frequencies from 1000 Hz to 5000 Hz, its bandwidth is $5000 - 1000 = 4000$ Hz.

Bandwidth in bits per second — the number of bits a channel can carry per second.

2.2.2 Bit Rate vs Baud Rate

Bit rate = number of *bits* sent per second (bps).
Baud rate (signal rate) = number of *signal units / symbols* sent per second (baud).

$$\boxed{\text{Bit rate} = \text{Baud rate} \times \log_2 L}$$

where L is the number of signal levels. Equivalently, $\text{Baud} = \dfrac{\text{Bit rate}}{\log_2 L}$.

Point	Bit Rate	Baud Rate
Measures	Bits per second	Signal units per second
Unit	bps	baud
Determines	Data transfer speed	Bandwidth required
Relation	Bit rate $\geq$ baud rate	Baud rate $\leq$ bit rate

Example 2.2 — Bit Rate from Baud Rate

A signal carries 4 bits per signal element and is transmitted at 2000 baud. Find the bit rate. Here $\log_2 L = 4$ (that is, $L = 16$ levels).

$$\text{Bit rate} = 2000 \times 4 = \mathbf{8000\ bps} = 8\ \text{kbps}$$

Example 2.3 — Baud Rate from Bit Rate

An analog signal carries 4 bits per signal element. If 1000 signal elements are sent per second, find the bit rate. Then, if the bit rate is 6000 bps and each signal element carries 3 bits, find the baud rate.
Part (a): Bit rate $= 1000 \times 4 = \mathbf{4000\ bps}$
Part (b): Baud rate $= \dfrac{6000}{3} = \mathbf{2000\ baud}$

Common Mistake

Bit rate and baud rate are *equal only when each signal element carries exactly one bit* ($L = 2$, so $\log_2 2 = 1$). Writing "bit rate = baud rate" as a general statement is a guaranteed mark loss.

2.2.3 Throughput, Latency and Bandwidth-Delay Product

Throughput — the *actual* measured rate at which data passes through the network. Bandwidth is the theoretical maximum; throughput is what you really get. Throughput is always $\leq$ bandwidth.

Latency (delay) — the total time for a message to travel from source to destination:

$$\text{Latency} = \text{Propagation time} + \text{Transmission time} + \text{Queuing time} + \text{Processing delay}$$

$$\text{Propagation time} = \frac{\text{Distance}}{\text{Propagation speed}} \qquad \text{Transmission time} = \frac{\text{Message size}}{\text{Bandwidth}}$$

Point	Bandwidth	Throughput
Meaning	Maximum possible capacity	Actual delivered rate
Nature	Theoretical	Practical, measured
Affected by	Physical medium	Traffic, errors, congestion
Value	Always $\geq$ throughput	Always $\leq$ bandwidth

Easy Way to Remember

A water pipe is the classic picture. **Bandwidth** = the width of the pipe. **Throughput** = how much water actually comes out — which is less, because of bends, leaks and other users.

Example 2.4 — Throughput

A network with bandwidth 10 Mbps can pass only an average of 12,000 frames per minute, each frame carrying 10,000 bits. What is the throughput?

Step 1 — Total bits per minute.

$$12{,}000 \times 10{,}000 = 120{,}000{,}000 \text{ bits per minute}$$

Step 2 — Convert to per second.

$$\text{Throughput} = \frac{120{,}000{,}000}{60} = 2{,}000{,}000 \text{ bps} = \mathbf{2\ Mbps}$$

The throughput is only one-fifth of the 10 Mbps bandwidth.

2.3 Transmission Impairments

Signals travel through imperfect media, so the signal that arrives is not the signal that was sent. Three impairments are asked about.

2.3.1 Attenuation

Loss of signal strength (energy) as the signal travels through the medium. Longer distance means weaker signal. The remedy is an **amplifier** (for analog) or a **repeater** (for digital).

Attenuation is measured in **decibels (dB)**:

$$\text{dB} = 10 \log_{10} \frac{P_2}{P_1}$$

where P_1 is the input power and P_2 the output power.

A *negative* dB value means the signal was attenuated (lost power). A *positive* value means it was amplified.

Example 2.5 — Attenuation in Decibels

A signal travels through a transmission medium and its power is reduced to one-half. Calculate the attenuation in decibels.

$$P_2 = \tfrac{1}{2}P_1 \Rightarrow \frac{P_2}{P_1} = 0.5$$

$$\text{dB} = 10\log_{10}(0.5) = 10 \times (-0.3) = \mathbf{-3\ dB}$$

Remember this result. A loss of 3 dB always means the power has halved. A gain of 3 dB always means the power has doubled.

Example 2.6 — Cascaded dB

A signal passes through three points. From point 1 to 2 it loses 3 dB, from 2 to 3 it is amplified by 7 dB, and from 3 to 4 it loses 3 dB. What is the overall gain or loss?
Decibels are logarithmic, so they simply **add**:

$$-3 + 7 - 3 = \mathbf{+1\ dB}$$

The signal is 1 dB stronger at point 4 than at point 1.

2.3.2 Distortion

The signal **changes its shape**. This happens in composite signals, where different frequency components travel at different speeds through the medium and therefore arrive with different delays. The received wave no longer looks like the transmitted wave.

2.3.3 Noise

Unwanted signals added to the original. Types:

- **Thermal noise** — random motion of electrons in the wire.
- **Induced noise** — from motors, appliances and other electrical machinery.
- **Crosstalk** — signal from one wire leaking into an adjacent wire (you hear another conversation on the phone).
- **Impulse noise** — a sudden high-energy spike, e.g. from lightning or a power surge.

Signal-to-Noise Ratio (SNR) measures how much stronger the signal is than the noise:

$$\text{SNR} = \frac{\text{average signal power}}{\text{average noise power}} \qquad \text{SNR}_{\text{dB}} = 10\log_{10}\text{SNR}$$

A *high* SNR is good. A *low* SNR means the signal is buried in noise.

2.4 Channel Capacity: Nyquist and Shannon

This is the most commonly asked numerical in this chapter after subnetting.

2.4.1 Nyquist Formula (Noiseless Channel)

$$\boxed{C = 2 \times B \times \log_2 L}$$

where C = capacity in bps, B = bandwidth in Hz, L = number of signal levels.

2.4.2 Shannon Capacity (Noisy Channel)

$$\boxed{C = B \times \log_2(1 + \text{SNR})}$$

where SNR is the plain ratio, *not* in decibels.

Which Formula to Use

If the question mentions **signal levels** → use **Nyquist**.
If the question mentions **noise or SNR** → use **Shannon**.
If the question gives **both** → calculate both and take the **lower** value as the practical limit, then use Nyquist to find the required number of levels.

Example 2.7 — Nyquist

A noiseless channel has a bandwidth of 3000 Hz transmitting a signal with 2 signal levels. What is the maximum bit rate?

$$C = 2 \times 3000 \times \log_2 2 = 2 \times 3000 \times 1 = \mathbf{6000\ bps}$$

If instead 4 signal levels are used:

$$C = 2 \times 3000 \times \log_2 4 = 2 \times 3000 \times 2 = \mathbf{12{,}000\ bps}$$

Doubling the levels from 2 to 4 doubled the capacity, because $\log_2 4 = 2$.

Example 2.8 — Shannon

Calculate the theoretical highest bit rate of a telephone line with bandwidth 3000 Hz and an SNR of 3162.

$$C = 3000 \times \log_2(1 + 3162) = 3000 \times \log_2 3163$$

Now, $\log_2 3163 = \dfrac{\log_{10} 3163}{\log_{10} 2} = \dfrac{3.5001}{0.3010} \approx 11.62$

$$C = 3000 \times 11.62 \approx \mathbf{34{,}860\ bps} \approx 35\ \text{kbps}$$

This is exactly why old dial-up modems topped out near 33.6 kbps — it is a hard physical limit, not a design flaw.

Example 2.9 — Using Both Formulas

A channel has a bandwidth of 1 MHz and an SNR of 63. What is the appropriate bit rate, and how many signal levels are needed?
Step 1 — Shannon gives the upper limit.

$$C = 10^6 \times \log_2(1 + 63) = 10^6 \times \log_2 64 = 10^6 \times 6 = 6\ \text{Mbps}$$

Step 2 — Use Nyquist to find the levels needed for 6 Mbps.

$$6 \times 10^6 = 2 \times 10^6 \times \log_2 L$$

$$\log_2 L = 3 \Rightarrow L = 2^3 = \mathbf{8\ levels}$$

Answer: bit rate 6 Mbps using 8 signal levels.

Log Values Worth Memorising

$\log_2 2 = 1$, $\log_2 4 = 2$, $\log_2 8 = 3$, $\log_2 16 = 4$, $\log_2 32 = 5$, $\log_2 64 = 6$, $\log_2 128 = 7$, $\log_2 256 = 8$, $\log_2 1024 = 10$.

If the number is not a power of 2, use $\log_2 x = \dfrac{\log_{10} x}{0.3010}$.

CHAPTER 3

Digital and Analog Transmission

3.1 Line Coding

Asked In

> *What is line coding? What are the different line coding techniques?* — SPCBL AME 2021

Line coding is the process of converting **digital data into a digital signal**. The sender encodes bits into a signal; the receiver decodes the signal back into bits.

3.1.1 Characteristics of a Good Line Code

- **Self-synchronisation** — the receiver can recover the clock from the signal itself, so it never drifts out of step.
- **No DC component** — the average voltage should be zero, so the signal can pass through transformers and long cables.
- **Low bandwidth requirement** and **built-in error detection** are also desirable.

3.1.2 Categories of Line Coding

Category	Schemes	How It Works
Unipolar	NRZ	One non-zero voltage level; 0 is zero volts. Has a strong DC component.
Polar	NRZ-L, NRZ-I, RZ, Manchester, Differential Manchester	Uses both positive and negative voltages.
Bipolar	AMI, Pseudoternary	Three levels: positive, zero, negative.
Multilevel	2B1Q, 8B6T, 4D-PAM5	Encodes m data bits into n signal elements.
Multitransition	MLT-3	Transition-based, reduces required bandwidth.

3.1.3 The Schemes You Must Be Able to Draw

NRZ-L (Non-Return-to-Zero, Level) — the *voltage level* determines the bit. Typically 0 = positive voltage, 1 = negative voltage.

NRZ-I (Non-Return-to-Zero, Invert) — the *presence of a transition* at the start of the bit period means 1; no transition means 0.

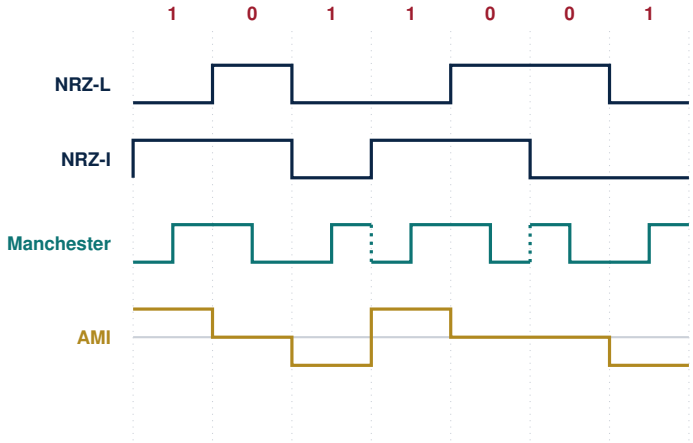
RZ (Return-to-Zero) — uses three values. The signal returns to zero in the middle of every bit. Good synchronisation, but needs double the bandwidth.

Manchester — combines RZ and NRZ-L. There is *always* a transition in the middle of every bit. Low-to-high = 1, high-to-low = 0 (IEEE convention). Self-synchronising, no DC component — this is why classic 10 Mbps Ethernet used it.

Differential Manchester — always a mid-bit transition for synchronisation, and the bit value is decided by whether there is *also* a transition at the *start* of the bit: transition at start = 0, no

transition at start = 1. Used in Token Ring.

AMI (Alternate Mark Inversion) — 0 is zero voltage; 1 alternates between positive and negative. This alternation removes the DC component.



Easy Way to Remember

Manchester in one line: "*Always change in the middle.*" The middle transition gives you the clock for free, which is why it never loses synchronisation — at the cost of needing twice the bandwidth.

3.2 Block Coding and Scrambling

Block coding converts a block of m bits into a block of n bits, where $n > m$. It is written as `mB/nB`. The extra bits provide redundancy for synchronisation and error detection.

Examples: `4B/5B` (used in 100BaseTX Fast Ethernet), `8B/10B` (Gigabit Ethernet), `8B/6T`.

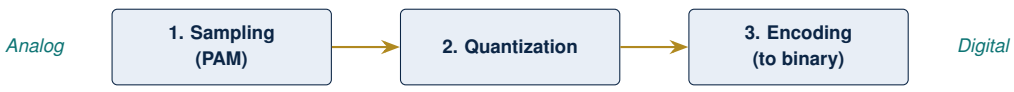
Scrambling replaces long sequences of zeros with a special pattern so that synchronisation is not lost. Two schemes: **B8ZS** (used in North America) and **HDB3** (used in Europe and Japan).

3.3 Pulse Code Modulation (PCM)

Asked In

What is modulation? Describe the pulse code modulation (PCM) technique. — BPSC AE 2022

PCM converts an **analog signal into digital data**. It is what happens inside every digital telephone system and every audio recording. There are three steps.



1. **Sampling** — measure the amplitude of the analog signal at regular intervals. The result is a series of pulses (Pulse Amplitude Modulation, PAM).
2. **Quantization** — round each sampled amplitude to the nearest of a fixed set of levels. This step introduces a small permanent error called **quantization noise**.
3. **Encoding** — convert each quantized level into a binary code.

3.3.1 Nyquist Sampling Theorem

$$\boxed{f_s \geq 2 \times f_{\max}}$$

The sampling rate must be *at least twice the highest frequency* present in the signal.

$$\text{Bit rate} = \text{sampling rate} \times \text{bits per sample}$$

Example 3.1 — PCM Bit Rate for a Telephone Channel

A telephone channel carries frequencies up to 4000 Hz. Each sample is quantized into 256 levels. Find the required sampling rate and the resulting bit rate.

Step 1 — Sampling rate.

$$f_s = 2 \times 4000 = 8000 \text{ samples per second}$$

Step 2 — Bits per sample.

$$256 = 2^8 \Rightarrow \log_2 256 = 8 \text{ bits per sample}$$

Step 3 — Bit rate.

$$\text{Bit rate} = 8000 \times 8 = \mathbf{64{,}000 \text{ bps}} = 64 \text{ kbps}$$

This 64 kbps figure is the standard digital voice channel (DS-0) used worldwide.

Example 3.2 — CD Audio

Audio CDs sample at 44,100 Hz using 16 bits per sample in stereo (2 channels). Find the bit rate.

$$44{,}100 \times 16 \times 2 = 1{,}411{,}200 \text{ bps} \approx \mathbf{1.41 \text{ Mbps}}$$

Common Mistake

Delta Modulation (DM) is a simplified alternative to PCM: instead of sending the full amplitude, it sends a single bit indicating only whether the signal went *up* or *down* since the last sample. It is simpler and cheaper but less accurate.

3.4 Digital-to-Analog Conversion (Modulation)

Asked In

What is modulation? Why is it necessary? — BPSC AE 2022

Modulation is the process of changing one or more characteristics of a high-frequency **carrier signal** in proportion to a **message signal**.

3.4.1 Why Modulation Is Necessary

- **Practical antenna size.** Antenna length must be about $\lambda/4$. A 3 kHz voice signal would need an antenna kilometres long. Raising the frequency shrinks the antenna to a usable size.
- **Long-distance transmission.** High-frequency signals travel much further.
- **Multiplexing.** Many signals can share one medium by being shifted to different carrier frequencies.
- **Reduced noise and interference.**

3.4.2 Digital-to-Analog Schemes

Scheme	What Changes	Description
ASK (Amplitude Shift Keying)	Amplitude	Two different amplitudes for 0 and 1. Simplest, but very sensitive to noise.
FSK (Frequency Shift Keying)	Frequency	Two different frequencies for 0 and 1. Noise-resistant, needs more bandwidth.
PSK (Phase Shift Keying)	Phase	Two or more phase shifts. Better noise immunity than ASK; used in Wi-Fi and Bluetooth.
QAM (Quadrature Amplitude Modulation)	Amplitude + Phase	Combines ASK and PSK. Highest data rate — used in cable modems, DSL, Wi-Fi, LTE.

Easy Way to Remember

A-F-P-Q. **A**SK changes **A**mplitude. **F**SK changes **F**requency. **P**SK changes **P**hase. **Q**AM changes **Q**uantity of both (amplitude *and* phase). The first letter tells you the answer.

3.4.3 Constellation Diagrams and Bits per Symbol

In QAM, the number of bits carried per symbol is $\log_2 L$ where L is the number of points in the constellation.

Example 3.3 — QAM Bit Rate

A 64-QAM system transmits at 2400 baud. Find the bit rate.

$$\log_2 64 = 6 \text{ bits per symbol}$$

$$\text{Bit rate} = 2400 \times 6 = \mathbf{14{,}400 \text{ bps}}$$

3.4.4 Analog-to-Analog Modulation

When the message signal is itself analog (as in radio broadcasting):

- **AM (Amplitude Modulation)** — carrier amplitude varies with the message. Used in AM radio. Bandwidth $= 2\times$ message bandwidth.
- **FM (Frequency Modulation)** — carrier frequency varies. Used in FM radio. Better quality, more bandwidth. Bandwidth $= 10\times$ message bandwidth.
- **PM (Phase Modulation)** — carrier phase varies.

Modem

A **modem** (**mo**dulator–**dem**odulator) converts digital data from a computer into an analog signal for transmission (modulation) and converts incoming analog signals back into digital data (demodulation). A **codec** does the opposite job: analog voice $\rightarrow$ digital.

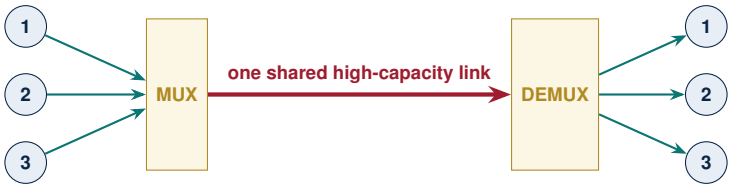
CHAPTER 4

Multiplexing and Transmission Media

4.1 Multiplexing

Multiplexing is the technique of sending **multiple signals over a single shared link** at the same time. The device that combines the signals is a **multiplexer (MUX)**; the device that separates them at the far end is a **demultiplexer (DEMUX)**.

Why multiplex? Because a single link usually has far more capacity than any one user needs. Sharing it reduces cost dramatically.



4.1.1 Frequency Division Multiplexing (FDM)

Analog technique. The bandwidth of the link is divided into several **frequency bands**, and each signal is modulated onto a different carrier frequency. Unused **guard bands** are placed between channels to prevent overlap.

Used in: radio and TV broadcasting, first-generation mobile phones, cable TV.

4.1.2 Time Division Multiplexing (TDM)

Digital technique. The link is divided in **time**, not frequency. Each connection gets the entire bandwidth, but only for a short repeating **time slot**.

- **Synchronous TDM** — each device gets a fixed slot in every frame, whether or not it has data. Empty slots are wasted.
- **Statistical (asynchronous) TDM** — slots are allocated *on demand* only to devices that actually have data. Much more efficient, but each slot must carry an address.

4.1.3 Wavelength Division Multiplexing (WDM)

FDM applied to **fibre-optic cable**, where the different "frequencies" are different **wavelengths (colours) of light**. A prism or diffraction grating combines and separates them. **DWDM** (Dense WDM) packs many closely spaced wavelengths onto one fibre.

Point	FDM	TDM	WDM
Signal type	Analog	Digital	Optical
Divides	Frequency	Time	Wavelength
Sharing	All transmit at once, different frequencies	Take turns in time slots	All transmit at once, different light colours
Guard bands	Required	Guard time	Required
Medium	Radio, coaxial	Any	Fibre optic only

Easy Way to Remember

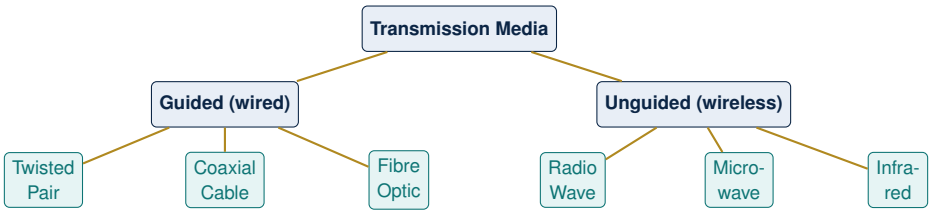
Classroom analogy. **FDM** = everyone speaks at once but in different languages, and each listener understands only one. **TDM** = everyone speaks the same language, but takes turns for a few seconds each. **WDM** = FDM done with light instead of radio.

4.1.4 Spread Spectrum

Deliberately spreads a signal over a *wider* bandwidth than necessary, to resist jamming and eavesdropping.

- **FHSS (Frequency Hopping Spread Spectrum)** — the carrier hops between many frequencies in a pseudo-random sequence known to both ends. Used in Bluetooth.
- **DSSS (Direct Sequence Spread Spectrum)** — each data bit is replaced by a longer code sequence (chip code). Used in Wi-Fi and CDMA.

4.2 Transmission Media



4.2.1 Twisted Pair Cable

Two insulated copper wires twisted together. **Twisting cancels out noise**, because both wires pick up the same interference and the receiver takes the difference.

- **UTP (Unshielded Twisted Pair)** — cheapest, easiest to install, most common in LANs. Connector: **RJ-45**.
- **STP (Shielded Twisted Pair)** — has a metal foil shield around the pairs; better noise immunity, more expensive, harder to install.

Asked In

Distinguish between Cat5 and Cat6. — recurring in bank IT officer papers

Category	Bandwidth	Data Rate	Max Length	Typical Use
Cat 3	16 MHz	10 Mbps	100 m	Telephone
Cat 5	100 MHz	100 Mbps	100 m	Fast Ethernet
Cat 5e	100 MHz	1 Gbps	100 m	Gigabit Ethernet
Cat 6	250 MHz	1 Gbps (10 Gbps to 55 m)	100 m	Gigabit / 10G short runs
Cat 6a	500 MHz	10 Gbps	100 m	10 Gigabit Ethernet
Cat 7	600 MHz	10 Gbps	100 m	Data centres

Cat5 vs Cat6 — The Short Answer

Cat6 has **more twists per inch**, a **thicker conductor** and often a **plastic spline** separating the pairs. Result: 250 MHz bandwidth instead of 100 MHz, far less crosstalk, and support for 10 Gbps over short distances. Cat6 is backward compatible with Cat5/5e.

4.2.2 Coaxial Cable

A central copper conductor, surrounded by insulation, then a braided metal shield, then an outer plastic jacket. The shield gives it much better noise immunity and higher bandwidth than twisted pair.

Used in: cable television, older Ethernet (10Base2 "Thinnet", 10Base5 "Thicknet"). Connector: **BNC**.

4.2.3 Fibre-Optic Cable

Transmits data as **pulses of light** through a glass or plastic core, using the principle of **total internal reflection**. The core has a higher refractive index than the surrounding cladding, so light bounces along inside instead of escaping.

- **Single-mode fibre** — very thin core (about 9 μm), one straight light path, uses a laser source. Very long distance (up to 100 km without repeater), highest bandwidth, most expensive.
- **Multi-mode fibre** — thicker core (50–62.5 μm), many light paths, uses an LED source. Shorter distance (up to about 2 km), cheaper.

Advantages: enormous bandwidth, immune to electromagnetic interference, very low attenuation, lightweight, highly secure (hard to tap without detection).
Disadvantages: expensive, fragile, requires skilled installation and specialised splicing equipment, unidirectional (needs two fibres for full duplex).

Point	Twisted Pair	Coaxial	Fibre Optic
Signal carried	Electrical	Electrical	Light
Bandwidth	Low	Medium	Very high
Data rate	Up to 10 Gbps	Up to 1 Gbps	Terabits per second
Attenuation	High	Medium	Very low
Noise immunity	Poor	Better	Immune (EMI-proof)
Distance	100 m	Few hundred metres	Up to 100 km
Cost	Cheapest	Moderate	Most expensive
Security	Easy to tap	Easy to tap	Very hard to tap

4.2.4 Unguided (Wireless) Media

Type	Frequency	Characteristics and Uses
Radio wave	3 kHz – 1 GHz	**Omnidirectional** — travels in all directions, penetrates walls, can bounce off the ionosphere for long distance. AM/FM radio, television, cordless phones.
Microwave	1 GHz – 300 GHz	**Unidirectional** — needs line of sight, uses parabolic dish antennas, cannot penetrate walls well. Satellite links, cellular backhaul, Wi-Fi.
Infrared	300 GHz – 400 THz	Very short range, cannot pass through walls (which makes it secure and non-interfering between rooms). TV remote controls, IrDA.

Asked In

Distinguish between microwave and radio wave. — past written papers

Radio vs Microwave — One Line Each

Radio waves are **omnidirectional** (broadcast in every direction, no dish needed, pass through walls). **Microwaves** are **unidirectional** (focused beam, need a dish and clear line of sight, blocked by walls and hills).

4.2.5 Satellite Communication

- **GEO (Geostationary Earth Orbit)** — 35,786 km. Appears fixed above one point. Only 3 satellites cover the whole earth. High propagation delay ($\approx$ 250 ms one way).
- **MEO (Medium Earth Orbit)** — 5,000–15,000 km. GPS satellites live here.
- **LEO (Low Earth Orbit)** — 500–2,000 km. Low delay, but needs many satellites (constellations such as Starlink and Iridium).

CHAPTER 5

Switching and Network Topologies

5.1 Switching Techniques

Asked In

Discuss circuit switching and packet switching with appropriate diagram. — BPSC
Why is packet switching suitable for digital data transmission? — BPSC AP 2022
What is circuit switch and packet switch? Which is better and why? — BCS (Cadre) 2017

5.1.1 Circuit Switching

A **dedicated physical path** is established between sender and receiver *before* any data flows, and it stays reserved for the whole session. The classic example is the old telephone network.

Three phases: **setup** → **data transfer** → **teardown**.

Advantages: guaranteed bandwidth, constant and predictable delay, no packet ordering problem, ideal for continuous voice.
Disadvantages: the path is reserved even when idle (very wasteful for bursty computer data), long setup time, and if the path breaks the whole call drops.

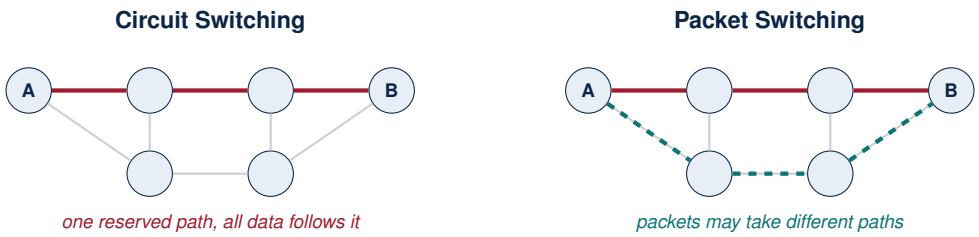
5.1.2 Packet Switching

Data is broken into **packets**, each carrying a header with source and destination addresses. Packets are forwarded hop by hop and share links with everybody else's traffic. No dedicated path is reserved.

- **Datagram approach (connectionless)** — each packet is routed independently and may take a different path. Packets can arrive out of order or be lost. This is how **IP** works.
- **Virtual circuit approach (connection-oriented)** — a logical path is set up first and all packets follow it in order, but the link is still shared. Used by Frame Relay and ATM.

5.1.3 Message Switching

The whole message is sent to an intermediate node, **stored completely**, then forwarded when a link is free (**store-and-forward**). No length limit on messages, but delay is very high. Now obsolete except in email-style systems.



Point	Circuit Switching	Packet Switching
Path	Dedicated, reserved	Shared, no reservation
Setup phase	Required before data	Not required (datagram)
Bandwidth	Fixed, guaranteed	Dynamic, on demand
Efficiency	Low — idle time wasted	High — links always usable
Delay	Fixed, predictable	Variable (queuing)
Data unit	Continuous bit stream	Packets
Store-and-forward	No	Yes
On failure	Call drops	Packets reroute around the fault
Best for	Voice, real-time	Bursty computer data
Example	Traditional telephone (PSTN)	The Internet, IP networks

Why Packet Switching Suits Digital Data

Computer traffic is **bursty** — a short download, then a long silence. Circuit switching would keep the whole line reserved during the silence, wasting it. Packet switching lets other users fill those gaps, so the same physical link serves many more users. It is also **fault tolerant**: if a router fails, packets simply route around it.

Easy Way to Remember

Circuit switching = booking a whole taxi for the day; it waits outside even while you are inside a meeting. **Packet switching** = taking buses; you share with others and may change route, but nothing sits idle.

5.2 Types of Network by Size

Type	Full Name	Coverage	Example
PAN	Personal Area Network	1–10 m	Bluetooth headset, smartwatch
LAN	Local Area Network	Building / campus	Office or university network
CAN	Campus Area Network	Few km	University campus
MAN	Metropolitan Area Network	City-wide	Cable TV network, city Wi-Fi
WAN	Wide Area Network	Country / global	The Internet, bank branch network

Point	LAN	WAN
Coverage	Small (one building)	Very large (countries)
Speed	High (100 Mbps – 10 Gbps)	Comparatively lower
Ownership	Private (single organisation)	Often public / leased
Error rate	Low	Higher
Cost	Low	High
Technology	Ethernet, Wi-Fi	Leased lines, MPLS, satellite

5.3 Client-Server vs Peer-to-Peer

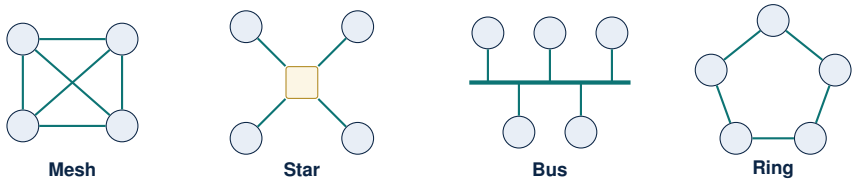
Asked In

Differentiate between client-server network and peer-to-peer network. — PSC Lecturer 2017

Point	Client-Server	Peer-to-Peer (P2P)
Roles	Dedicated server, many clients	Every node is both client and server
Data storage	Centralised on the server	Distributed on each node
Security	Strong, centrally managed	Weak, each user manages own
Cost	High (server hardware, admin)	Low
Scalability	Scales well with a strong server	Performance drops as nodes grow
Backup	Centralised, easy	Each node separately, hard
Failure	Server failure stops everything	No single point of failure
Example	Web, email, banking systems	BitTorrent, small home networks

5.4 Network Topologies

Topology is the physical or logical arrangement of devices on a network.



5.4.1 Mesh

Every device has a dedicated point-to-point link to every other device. Number of links needed: $\frac{n(n-1)}{2}$, and each device needs $n-1$ ports.

Advantages: no traffic contention, very robust (one link failure does not stop the network), privacy, easy fault identification.
Disadvantages: enormous cabling and port cost; impractical beyond small n.

Example 5.1 — Mesh Links

How many cables are needed to connect 8 devices in a full mesh?

$$\frac{n(n-1)}{2} = \frac{8 \times 7}{2} = \mathbf{28 \text{ links}}$$

And each device needs $8 - 1 = 7$ ports, so 56 ports in total.

5.4.2 Star

Every device connects to a **central hub or switch**. There is no direct device-to-device link.

Advantages: cheap (only n cables), easy to install and reconfigure, one cable failure affects only one device, easy fault isolation.
Disadvantage: the central hub is a **single point of failure** — if it dies, the entire network dies.

5.4.3 Bus

All devices tap into a single **backbone cable**, which is terminated at both ends.

Advantages: least cable, easy to install, cheap.
Disadvantages: a break in the backbone kills the whole network; difficult fault isolation; heavy collisions as devices increase.

5.4.4 Ring

Each device connects to exactly two neighbours, forming a closed loop. Data travels in one direction, and each device acts as a repeater.

Advantages: easy to install, each device regenerates the signal so distances are longer, orderly access (no collisions with token passing).
Disadvantage: a single break breaks the ring (solved by **dual ring**).

5.4.5 Hybrid and Tree

A combination — for example, a star-bus in which several star hubs are joined to one bus backbone. Most real corporate networks are hybrid.

Topology	Cables Needed	Failure Impact	Cost	Fault Isolation
Mesh	$n(n-1)/2$	Minimal	Very high	Easy
Star	n	Hub failure = total	Low	Easy
Bus	1 backbone	Backbone break = total	Lowest	Difficult
Ring	n	One break = total	Low	Moderate

Mesh vs Star — The Comparison Most Often Asked

Mesh: dedicated links between every pair, $n(n-1)/2$ cables, no central device, extremely reliable, extremely expensive.
Star: all links go to one central hub, only n cables, cheap and easy to expand, but the hub is a single point of failure.

CHAPTER 6

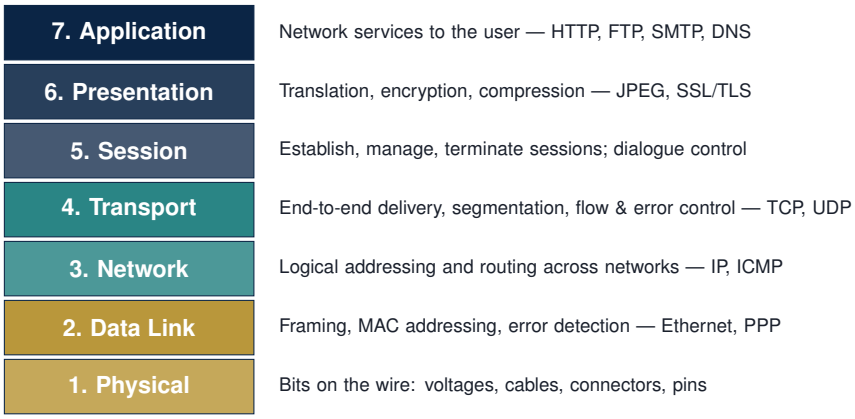
OSI and TCP/IP Reference Models

6.1 Why Layered Models Exist

A layered model breaks a huge, complex problem into small independent pieces. Each layer does one job, offers a service to the layer above it, and can be changed without disturbing the others.

6.2 The OSI Model — Seven Layers

The **Open Systems Interconnection (OSI)** model was defined by **ISO**. It is a reference framework, not an implemented protocol stack.



6.2.1 Layer by Layer

Layer 1 — Physical. Transmits raw bits over the medium. Deals with voltage levels, data rate, cable and connector specifications, and the physical topology.
Data unit: bit. Devices: hub, repeater, cable, connector.

Layer 2 — Data Link. Makes the raw physical link reliable between two directly connected nodes. Performs *framing*, *physical (MAC) addressing*, *error detection*, *flow control* and *access control*. Split into two sublayers: **LLC** (Logical Link Control) and **MAC** (Media Access Control).
Data unit: frame. Devices: switch, bridge, NIC.

Layer 3 — Network. Delivers packets from the source host to the destination host across multiple networks. Performs *logical addressing* (IP addresses) and *routing*.
Data unit: packet. Devices: router, layer-3 switch.

Layer 4 — Transport. Provides *process-to-process* delivery using *port numbers*. Performs segmentation and reassembly, connection control, end-to-end flow control and error control.
Data unit: segment (TCP) or datagram (UDP). Device: gateway.

Layer 5 — Session. Establishes, maintains and terminates sessions between applications. Handles *dialogue control* (who talks when) and *synchronisation* (checkpoints so a long transfer can resume rather than restart).

Layer 6 — Presentation. Concerned with the *syntax and semantics* of the data: *translation* (ASCII

↔ EBCDIC), *encryption/decryption*, and *compression*.

Layer 7 — Application. The layer the user actually touches. Provides network services to applications: file transfer, email, remote login, directory services.

Memorising the Seven Layers

Bottom to top (1→7):
Please **D**o **N**ot **T**hrow **S**ausage **P**izza **A**way
Physical, Data link, Network, Transport, Session, Presentation, Application

Top to bottom (7→1):
All **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing

The Data Unit at Each Layer — Very Frequently Asked

Application / Presentation / Session → **Data**
Transport → **Segment** (TCP) or **Datagram** (UDP)
Network → **Packet**
Data Link → **Frame**
Physical → **Bit**

6.3 Encapsulation

As data moves *down* the layers at the sender, each layer adds its own **header** (and the data link layer also adds a **trailer**). This is **encapsulation**. At the receiver the process runs in reverse (**decapsulation**), each layer stripping off its own header.



Frame = Eth Hdr + (IP Hdr + (TCP Hdr + Data)) + Eth Trailer

6.4 The TCP/IP Model

The practical model actually used on the Internet. It has **four layers** (some texts show five by splitting the Network Access layer).

TCP/IP Layer	OSI Equivalent	Protocols
Application	Layers 5, 6, 7	HTTP, FTP, SMTP, DNS, DHCP, SNMP, Telnet
Transport	Layer 4	TCP, UDP, SCTP
Internet (Network)	Layer 3	IP, ICMP, IGMP, ARP, RARP
Network Access (Link)	Layers 1, 2	Ethernet, Wi-Fi, PPP, Token Ring

Point	OSI Model	TCP/IP Model
Number of layers	7	4 (or 5)
Developed by	ISO	US Department of Defense / ARPANET
Nature	Theoretical reference model	Practical, implemented model
Order	Model came first, protocols later	Protocols came first, model described later
Layer coupling	Layers strictly independent	Layers less strictly separated
Transport	Connection-oriented only	Both connection-oriented (TCP) and connectionless (UDP)
Usage today	Teaching and troubleshooting reference	Actually runs the Internet

Common Mistake

A very common written-exam trap: *"Which OSI layer does a router work at?"*
Answer: the **Network layer (Layer 3)**.
Switch and bridge → **Data Link (Layer 2)**.
Hub and repeater → **Physical (Layer 1)**.
Gateway → can operate at **all seven layers**.

CHAPTER 7

Error Detection, Correction and Flow Control

7.1 Types of Error

Single-bit error — exactly one bit in the data unit changes from 0 to 1 or 1 to 0. Rare in serial transmission.

Burst error — two or more bits in the data unit change. The burst *length* is counted from the first corrupted bit to the last, and bits in between may be unaffected. This is the common case, because noise usually lasts longer than one bit period.

Easy Way to Remember

The faster the data rate, the more bits fit inside the same burst of noise. A 1 ms noise spike corrupts 10 bits at 10 kbps, but 1,000 bits at 1 Mbps. This is why high-speed links need strong error control.

7.2 Redundancy

All error detection works the same way: the sender adds extra bits (**redundant bits**) computed from the data. The receiver recomputes them and compares. If they disagree, an error occurred.

7.3 Parity Check

7.3.1 Simple Parity Check

One **parity bit** is added so that the total number of 1s becomes even (**even parity**) or odd (**odd parity**).

Example 7.1 — Even Parity

Data to send: `1011001`
Count the 1s: $1+0+1+1+0+0+1 = 4$ ones. That is already even, so the parity bit is `0`.
Transmitted: `10110010`

Now take data `1010001`. Number of 1s $= 3$ (odd), so the parity bit must be `1` to make it even.
Transmitted: `10100011`

Common Mistake

Simple parity detects **all odd numbers** of bit errors but **cannot detect an even number** of errors. If two bits flip, the parity is unchanged and the error passes undetected. It also cannot *correct* anything.

7.3.2 Two-Dimensional Parity Check

Data is arranged in a table of rows and columns. A parity bit is computed for every row *and* every column. This detects most burst errors and can correct a single-bit error (because the wrong row and wrong column intersect at the faulty bit).

7.4 Checksum

Used by TCP, UDP and IP. It works on blocks of n bits.

Checksum Procedure

At the sender: (1) divide the data into k blocks of n bits; (2) add all blocks using *one's complement* arithmetic; (3) complement the sum — this is the checksum; (4) send data + checksum.
At the receiver: (1) add all received blocks *including* the checksum; (2) complement the result; (3) if the result is **all zeros**, the data is accepted; otherwise it is discarded.

Example 7.2 — Checksum with 4-bit Blocks

Data blocks: `1001`, `1110`, `0011`
Sender side:

$$1001 + 1110 = 10111 \quad \text{(5 bits — carry out)}$$
$$\text{wrap the carry:} \quad 0111 + 1 = 1000$$
$$1000 + 0011 = 1011 \quad \text{(no carry)}$$

Sum $=$ `1011`. Complement it: **Checksum** $=$ `0100`.
Transmitted: `1001 1110 0011 0100`
Receiver side (no error):

$$1001 + 1110 + 0011 + 0100 = 1111$$

Complement of `1111` is `0000` $\rightarrow$ all zeros $\rightarrow$ **data accepted**.

7.5 Cyclic Redundancy Check (CRC)

The most powerful and most widely used detection method — Ethernet, Wi-Fi and almost every link layer uses it. It is based on **binary division using XOR** (modulo-2 arithmetic, no borrows or carries).

CRC Procedure

Sender: (1) the divisor (*generator polynomial*) has n bits; (2) append $n-1$ zeros to the data; (3) divide by the generator using XOR; (4) the remainder ($n-1$ bits) is the **CRC**; (5) replace the appended zeros with the CRC and transmit.
Receiver: divide the whole received bit string by the same generator.
Remainder $=$ **0** $\rightarrow$ no error. **Remainder** $\neq$ **0** $\rightarrow$ error, discard the frame.

XOR Rule

$0 \oplus 0 = 0, \quad 1 \oplus 1 = 0, \quad 0 \oplus 1 = 1, \quad 1 \oplus 0 = 1.$
Same bits give 0, different bits give 1. There are no carries and no borrows — this is what makes modulo-2 division easier than ordinary division.

Example 7.3 — CRC Step by Step

Data: `100100` **Generator:** `1101` (4 bits, so $n - 1 = 3$)
Step 1 — Append 3 zeros: `100100` **000** $\rightarrow$ `100100000`
Step 2 — Divide by 1101 using XOR:

```
100100000   ÷ 1101
1101
----
01000
  1101
  ----
  01010
    1101
    ----
    01110
      1101
      ----
       001
```

Step 3 — Remainder (CRC) = `001`
Step 4 — Transmitted frame: `100100` **001** $\rightarrow$ `100100001`

Check at the receiver: divide `100100001` by `1101`. The remainder comes out `000`, so the frame is accepted.

What CRC Detects

A well-chosen generator detects: all single-bit errors, all double-bit errors, any odd number of errors, and **all burst errors shorter than the length of the generator**. Standard CRC-32 (used in Ethernet) catches over 99.99% of all errors.

7.6 Hamming Code (Error Correction)

Parity, checksum and CRC only *detect* errors. **Hamming code** can also **correct** a single-bit error, because the redundant bits tell you exactly *which* bit is wrong.

7.6.1 How Many Redundant Bits?

$$\boxed{2^r \geq m + r + 1}$$

where m = number of data bits and r = number of redundant (parity) bits.

Example 7.4 — Number of Redundant Bits

How many parity bits are needed for 7 data bits?
Try $r = 3$: $2^3 = 8$ and $m + r + 1 = 7 + 3 + 1 = 11$. $8 \geq 11$ is *false*.
Try $r = 4$: $2^4 = 16$ and $7 + 4 + 1 = 12$. $16 \geq 12$ is *true*.
Answer: 4 redundant bits, giving an 11-bit codeword.

7.6.2 Placing and Computing the Parity Bits

Parity bits go in positions that are powers of 2: positions **1, 2, 4, 8, 16**, ... All other positions hold data bits.

Each parity bit checks the positions whose binary representation has a 1 in the corresponding place:

- P_1 (position 1) checks positions 1, 3, 5, 7, 9, 11 — *check 1, skip 1*
- P_2 (position 2) checks positions 2, 3, 6, 7, 10, 11 — *check 2, skip 2*
- P_4 (position 4) checks positions 4, 5, 6, 7, 12, 13, 14, 15 — *check 4, skip 4*
- P_8 (position 8) checks positions 8–15 — *check 8, skip 8*

Example 7.5 — Building a Hamming Code

Encode the 4-bit data `1011` using even parity.
Step 1 — Find r. $m = 4$. Try $r = 3$: $2^3 = 8 \geq 4 + 3 + 1 = 8$ ✓. So $r = 3$, total codeword length $= 7$.
Step 2 — Lay out the positions. Data bits `1011` fill positions 3, 5, 6, 7.

Position	1	2	3	4	5	6	7
Bit	P_1	P_2	1	P_4	0	1	1

Step 3 — Compute each parity bit (even parity).
P_1 checks positions 1, 3, 5, 7 → values P_1, 1, 0, 1.
Number of 1s among the data = 2 (even), so $P_1 = \mathbf{0}$.
P_2 checks positions 2, 3, 6, 7 → values P_2, 1, 1, 1.
Number of 1s = 3 (odd), so $P_2 = \mathbf{1}$.
P_4 checks positions 4, 5, 6, 7 → values P_4, 0, 1, 1.
Number of 1s = 2 (even), so $P_4 = \mathbf{0}$.
Step 4 — Final codeword: `0110011`

1	2	3	4	5	6	7
0	1	1	0	0	1	1

Example 7.6 — Detecting and Correcting an Error

The codeword `0110011` was sent, but `0110111` was received (bit 5 flipped).
Recompute each parity check on the *received* word:
C_1 (positions 1,3,5,7) $= 0, 1, 1, 1 \rightarrow$ three 1s, odd → **fails** → $C_1 = 1$
C_2 (positions 2,3,6,7) $= 1, 1, 1, 1 \rightarrow$ four 1s, even → passes → $C_2 = 0$
C_4 (positions 4,5,6,7) $= 0, 1, 1, 1 \rightarrow$ three 1s, odd → **fails** → $C_4 = 1$
Form the syndrome as $C_4C_2C_1 = 101_2 = \mathbf{5}$
Conclusion: bit position 5 is wrong. Flip it back from 1 to 0 and the original `0110011` is recovered.

Easy Way to Remember

The syndrome *is* the position number in binary. If the syndrome is 0, there is no error. That is the whole elegance of Hamming code — the failed checks spell out the address of the broken bit.

7.7 Flow Control and Error Control Protocols

Flow control stops a fast sender from overwhelming a slow receiver. **Error control** is flow control plus retransmission of damaged or lost frames — this is called **ARQ (Automatic Repeat reQuest)**.

7.7.1 Stop-and-Wait ARQ

Send one frame, then *wait* for its acknowledgement before sending the next. A timer runs; if the ACK does not arrive before it expires, the frame is retransmitted.

Simple, but very inefficient on long or fast links, because the sender is idle for one full round-trip time after every single frame.

7.7.2 Go-Back-N ARQ

The sender may transmit up to N frames (the **window**) without waiting. If frame k is damaged, the receiver discards it *and every frame after it*, and the sender retransmits frame k onwards.

Window size: $2^m - 1$ where m is the number of bits in the sequence number.

7.7.3 Selective Repeat ARQ

Only the specific damaged frame is retransmitted. The receiver buffers out-of-order frames and reorders them. More efficient in bandwidth but needs more memory and more complex logic.

Window size: 2^{m-1}.

Point	Stop-and-Wait	Go-Back-N	Selective Repeat
Window size	1	$2^m - 1$	2^{m-1}
Retransmits	The one frame	The bad frame and all after	Only the bad frame
Receiver buffer	1 frame	1 frame	Many frames
Efficiency	Lowest	Medium	Highest
Complexity	Simplest	Moderate	Most complex

Easy Way to Remember

Go-Back-N = a teacher who, on finding one wrong answer, makes you redo that question *and every question after it*. **Selective Repeat** = a teacher who makes you redo only the one wrong question.

CHAPTER 8

Multiple Access, Ethernet and Network Devices

8.1 Multiple Access Protocols

When many devices share one medium, something must decide who transmits when — otherwise signals collide.

8.1.1 ALOHA

Pure ALOHA — transmit whenever you have data. If a collision occurs, wait a random time and retry. Maximum efficiency only **18.4%**.

Slotted ALOHA — time is divided into slots and stations may transmit only at the start of a slot. This halves the vulnerable period, doubling efficiency to **36.8%**.

8.1.2 CSMA (Carrier Sense Multiple Access)

"**Listen before you talk**." A station senses the channel first and transmits only if it is idle.

Persistence methods:

- **1-persistent** — if idle, send immediately (probability 1). High collision chance.
- **Non-persistent** — if busy, wait a random time, then sense again. Fewer collisions but longer delay.
- **p-persistent** — used with slotted channels; if idle, send with probability p.

8.1.3 CSMA/CD — Collision Detection

"**Listen before and while you talk**." Used in **wired Ethernet**. If a collision is detected during transmission, the station immediately stops, sends a **jam signal**, and waits a random period determined by the **binary exponential backoff** algorithm before retrying.

8.1.4 CSMA/CA — Collision Avoidance

Used in **wireless (Wi-Fi, 802.11)**. In wireless a station cannot listen while transmitting (its own signal drowns everything else), so collisions cannot be *detected* — they must be *avoided*. It uses interframe spacing, a contention window with random backoff, and explicit ACKs. Optionally it uses **RTS/CTS** handshaking to solve the **hidden terminal problem**.

Point	CSMA/CD	CSMA/CA
Used in	Wired Ethernet (802.3)	Wireless LAN (802.11)
Approach	Detect collisions after they start	Avoid collisions before they happen
Action on collision	Stop, jam, backoff, retry	Collisions are not detected at all
Acknowledgement	Not used	Explicit ACK required
Extra mechanism	Binary exponential backoff	RTS/CTS, interframe spacing
Efficiency	Higher	Lower (more overhead)

Why Wireless Cannot Use CD

A wireless radio cannot transmit and receive on the same frequency at the same time — its own transmission is millions of times stronger than any incoming signal. Since it cannot *hear* a collision, it must *avoid* one. This single fact is the complete answer to "why CSMA/CA in wireless?"

8.1.5 Controlled Access

Token passing — a special frame (the **token**) circulates; only the station holding the token may transmit. Used in Token Ring (802.5) and FDDI. No collisions ever occur, and delay is predictable.

Polling — a primary station asks each secondary in turn whether it has data.

Reservation — stations reserve slots in advance.

8.2 Ethernet Standards

Asked In

Write short notes on 10Base2, 10Base5, 10BaseF, 10BaseT, 100BaseT and 1000BaseTX. — past written papers

The naming convention is: `[speed in Mbps] [Base/Broad] [medium or segment length]`.

Standard	Speed	Medium	Max Length	Topology
10Base5	10 Mbps	Thick coaxial	500 m	Bus
10Base2	10 Mbps	Thin coaxial	185 m	Bus
10BaseT	10 Mbps	UTP (Cat 3+)	100 m	Star
10BaseF	10 Mbps	Fibre optic	2000 m	Star
100BaseTX	100 Mbps	UTP Cat 5 (2 pairs)	100 m	Star
100BaseFX	100 Mbps	Fibre optic	2000 m	Star
1000BaseT	1 Gbps	UTP Cat 5e (4 pairs)	100 m	Star
1000BaseTX	1 Gbps	UTP Cat 6	100 m	Star
1000BaseLX	1 Gbps	Single-mode fibre	5000 m	Star

Easy Way to Remember

"**Base**" means **baseband** — the whole cable carries one digital signal. "**Broad**" means **broadband** — the cable is divided by FDM into many analog channels. In `10Base5` the 5 means 500 m; in `10BaseT` the T means twisted pair.

8.2.1 Baseband vs Broadband

Point	Baseband	Broadband
Signal	Digital	Analog
Channels	One, uses whole bandwidth	Many, via FDM
Direction	Bidirectional	Unidirectional per channel
Multiplexing	TDM	FDM
Distance	Short	Long
Example	Ethernet LAN	Cable TV, DSL

8.3 Collision Domain and Broadcast Domain

Asked In

Differentiate between collision domain and broadcast domain. — Rupali Bank ANE 2023
Explain the terms domains, bandwidth, broadcast and multicast. — MMAP 2017

Collision domain — a network segment where simultaneous transmissions will collide. **Broadcast domain** — the set of devices that will all receive a broadcast frame.

Device	Collision Domains	Broadcast Domains
Hub	1 (all ports share one)	1
Bridge / Switch	1 per port	1 (unless VLANs are configured)
Router	1 per port	1 per port (routers block broadcasts)

The One-Line Answer

A **switch breaks up collision domains**.
A **router breaks up broadcast domains**.
A **VLAN** also breaks up broadcast domains, but logically instead of physically.

8.3.1 Unicast, Multicast, Broadcast

- **Unicast** — one sender to *one* specific receiver.
- **Multicast** — one sender to a *group* of interested receivers (IPTV, video conferencing). IPv4 multicast uses Class D addresses.
- **Broadcast** — one sender to *every* device on the network. MAC broadcast address is `FF:FF:FF:FF:FF:FF`. *IPv6 has no broadcast* — it uses multicast instead.
- **Anycast** — one sender to the *nearest* member of a group (used by DNS root servers and CDNs).

8.4 Networking Devices

Device	OSI Layer	Function
Repeater	1 Physical	Regenerates and amplifies a weak signal to extend distance. No filtering.
Hub	1 Physical	A multiport repeater. Forwards incoming data out of *every* other port. One collision domain, half duplex. Now obsolete.
Bridge	2 Data Link	Connects two LAN segments and filters traffic using MAC addresses.
Switch	2 Data Link	A multiport bridge. Learns MAC addresses and forwards frames *only* to the correct port. Each port is its own collision domain; supports full duplex and VLANs.
Router	3 Network	Connects different networks. Forwards packets using IP addresses and a routing table. Blocks broadcasts.
Layer-3 switch	3 Network	A switch with routing built in — much faster than a traditional router for internal traffic.
Gateway	All 7	Connects two networks that use *different protocols*, translating between them.
NIC	1 & 2	Network Interface Card — provides the physical connection and holds the permanent MAC address.
Modem	1 Physical	Modulates digital to analog and back for transmission over telephone or cable lines.
Access Point	2 Data Link	Lets wireless devices join a wired network.
Firewall	3, 4 & 7	Filters traffic according to security rules.

Point	Hub	Switch	Router
Layer	Physical (1)	Data Link (2)	Network (3)
Addressing	None	MAC address	IP address
Forwarding	To all ports	Only to correct port	Best path from routing table
Collision domain	One for all	One per port	One per port
Broadcast domain	One	One	One per port
Duplex	Half	Full	Full
Intelligence	None	Learns MAC table	Routing decisions

Easy Way to Remember

Hub shouts to everyone in the room. **Switch** whispers to exactly the right person. **Router** decides which room the message should go to next.

8.4.1 ARP and RARP

ARP (Address Resolution Protocol) — finds the *MAC address* when you know the *IP address*. It broadcasts "who has this IP?" and the owner replies with its MAC.

RARP (Reverse ARP) — finds the *IP address* when you know the *MAC address*. Now replaced by **BOOTP** and **DHCP**.

Easy Way to Remember

ARP: IP $\rightarrow$ MAC. **RARP: MAC $\rightarrow$ IP.**
ARP is the one you use every day; remember its direction and the reverse follows.

CHAPTER 9

IP Addressing and Subnetting

Asked In

This is the **highest-scoring chapter** in the whole syllabus. Recent appearances:
Convert the decimal IP 192.168.101.5 into binary. — BREB AP 2023
What is a subnet? What benefits will you get using subnets for this office? — Combined Bank SO(IT)
What is private IP range for class A, B and C with maximum hosts of each class? — BREB AP 2023
What is the motivation for moving from classful to classless addressing? — Sonali Bank
Find the IP address assignments for different departments/subnets. — BPSC AP

9.1 Binary Basics You Must Be Fluent In

Every subnetting question is really a binary question. Learn this one row and everything else follows.

128	64	32	16	8	4	2	1
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

9.1.1 Decimal to Binary — The Subtraction Method

Take the decimal number. Move left to right through 128, 64, 32, 16, 8, 4, 2, 1. If the value fits, write 1 and subtract it; if not, write 0.

Example 9.1 — Convert 192.168.101.5 to Binary (BREB AP 2023)

192: $192 - 128 = 64$ ✓(1); $64 - 64 = 0$ ✓(1); everything else 0 → `11000000`
168: $168 - 128 = 40$ (1); 64 too big (0); $40 - 32 = 8$ (1); 16 too big (0); $8 - 8 = 0$ (1); rest 0 → `10101000`
101: 128 too big (0); $101 - 64 = 37$ (1); $37 - 32 = 5$ (1); 16 too big (0); 8 too big (0); $5 - 4 = 1$ (1); 2 too big (0); $1 - 1 = 0$ (1) → `01100101`
5: $5 = 4 + 1$ → `00000101`
Answer: `11000000.10101000.01100101.00000101`

Values Worth Memorising Cold

2^1=2, 2^2=4, 2^3=8, 2^4=16, 2^5=32, 2^6=64, 2^7=128, 2^8=256, 2^9=512, 2^{10}=1024, 2^{16}=65,536, 2^{24}=16,777,216.
Also memorise the only eight legal subnet-mask octets:
0, 128, 192, 224, 240, 248, 252, 254, 255

9.2 IPv4 Address Structure

An IPv4 address is **32 bits**, written as four **octets** (bytes) in **dotted decimal notation**, each 0–255.

Every IP address has two parts: the **network ID** (which network) and the **host ID** (which device on that network).

9.3 Classful Addressing

Class	First Bits	First Octet	Default Mask	Networks	Hosts per Network
A	0	1 – 126	255.0.0.0 (/8)	126	16,777,214
B	10	128 – 191	255.255.0.0 (/16)	16,384	65,534
C	110	192 – 223	255.255.255.0 (/24)	2,097,152	254
D	1110	224 – 239	— (multicast)	—	—
E	1111	240 – 255	— (reserved)	—	—

Why the Host Counts End in 4 and Not 6

Hosts per network $= 2^h - 2$, where h is the number of host bits.
You subtract **2** because two addresses can never be given to a device:
- all host bits **0** → the **network address**
- all host bits **1** → the **broadcast address**

Common Mistake

127.x.x.x is missing from Class A on purpose. The entire 127 block is reserved for **loopback** (127.0.0.1 = localhost). That is why Class A runs 1–126, not 1–127.

9.3.1 Private IP Ranges

These addresses are not routable on the public Internet. They are reused inside every private network and translated by NAT.

Class	Private Range	CIDR	Number of Addresses
A	10.0.0.0 – 10.255.255.255	10.0.0.0/8	16,777,216
B	172.16.0.0 – 172.31.255.255	172.16.0.0/12	1,048,576
C	192.168.0.0 – 192.168.255.255	192.168.0.0/16	65,536

Easy Way to Remember

10 → one number, Class **A**.
172.16–31 → Class **B** (note: only 16 to 31, not the whole 172).
192.168 → Class **C**, the one on your home router.
Also remember **APIPA**: **169.254.x.x** is what Windows self-assigns when DHCP fails. Seeing it means "no DHCP server reachable".

9.4 Classless Addressing (CIDR)

Asked In

What is the primary motivation for moving from classful to classless IP addressing? — Sonali Bank

Why Classful Addressing Failed

Massive waste. An organisation with 300 computers was too big for a Class C (254 hosts) so it received a Class B — and wasted 65,234 addresses. There was no size in between.
Exhaustion. IPv4 has only about 4.3 billion addresses, and rigid classes burned through them far too fast.
Routing table explosion. Millions of separate class-based routes could not be aggregated.
CIDR (Classless Inter-Domain Routing) solved all three by letting the network/host boundary sit at *any* bit position, written as a **prefix length** such as /26. It also allows **route aggregation (supernetting)**, which shrinks routing tables.

9.4.1 Prefix Length and Subnet Mask

The **prefix length** /n means the first n bits are the network part and the remaining $32 - n$ bits are the host part.

Prefix	Subnet Mask	Host Bits	Block Size	Usable Hosts	Subnets in /24
/24	255.255.255.0	8	256	254	1
/25	255.255.255.128	7	128	126	2
/26	255.255.255.192	6	64	62	4
/27	255.255.255.224	5	32	30	8
/28	255.255.255.240	4	16	14	16
/29	255.255.255.248	3	8	6	32
/30	255.255.255.252	2	4	2	64

The Three Formulas That Solve Every Subnetting Question

$$\text{Number of subnets} = 2^s \quad \text{where } s = \text{bits borrowed from the host part}$$

$$\text{Usable hosts per subnet} = 2^h - 2 \quad \text{where } h = 32 - \text{prefix length}$$

$$\text{Block size (increment)} = 256 - (\text{the interesting octet of the mask})$$

The **interesting octet** is the last non-255 octet of the subnet mask. The **block size** tells you where every subnet starts.

9.5 Subnetting — The Five-Step Method

Use this exact sequence for every question and you will not make mistakes.



Example 9.2 — The Standard Question, Done Two Ways

Given: IP address `192.168.1.50`, subnet mask `255.255.255.240`.
Find the network address, broadcast address and valid host range.

METHOD 1 — Binary (always works, use this in the exam)
IP: `11000000.10101000.00000001.00110010`
Mask: `11111111.11111111.11111111.11110000`
*Network address = IP **AND** Mask* (AND gives 1 only when both bits are 1):

```
  11000000.10101000.00000001.0011 0010
  11111111.11111111.11111111.1111 0000
  ------------------------------------
  11000000.10101000.00000001.0011 0000  = 192.168.1.48
```

Broadcast address = network address with all host bits set to 1:

```
  11000000.10101000.00000001.0011 1111  = 192.168.1.63
```

METHOD 2 — Block Size (fast, use this to check your answer)
Block size $= 256 - 240 =$ **16**
Subnets increment by 16: `.0`, `.16`, `.32`, **`.48`**, `.64`, `.80`, ...
Our host `.50` falls in the block starting at **.48** and ending just before `.64`.

Network address	`192.168.1.48`
First usable host	`192.168.1.49`
Last usable host	`192.168.1.62`
Broadcast address	`192.168.1.63`
Usable hosts	$2^4 - 2 = \mathbf{14}$

Example 9.3 — A /27 Network

Given: `192.168.5.130/27`. Find all details.
Step 1 — Mask. /27 means 27 network bits, so the mask is
`11111111.11111111.11111111.11100000` = `255.255.255.224`
Step 2 — Block size. $256 - 224 =$ **32**
Step 3 — Subnet boundaries. `.0`, `.32`, `.64`, `.96`, **`.128`**, `.160`, `.192`, `.224`
Step 4 — Locate .130. It sits between `.128` and `.160`, so its network is `.128`.

Network address	`192.168.5.128`
First usable host	`192.168.5.129`
Last usable host	`192.168.5.158`
Broadcast address	`192.168.5.159`
Subnet mask	`255.255.255.224`
Usable hosts	$2^5 - 2 = \mathbf{30}$

Example 9.4 — "I Need N Subnets"

Question: You are given `192.168.10.0/24` and need **6 subnets**. Design them.
Step 1 — How many bits to borrow? We need $2^s \geq 6$.

$2^2 = 4$ (not enough), $2^3 = 8 \geq 6$ ✓ So borrow **3 bits**.
Step 2 — New prefix. $24 + 3 =$ **/27**, mask `255.255.255.224`
Step 3 — Hosts per subnet. Host bits $= 32 - 27 = 5$, so $2^5 - 2 =$ **30** hosts each.
Step 4 — Block size $= 256 - 224 = 32$.

Subnet	Network	First Host	Last Host	Broadcast
1	192.168.10.0	.1	.30	.31
2	192.168.10.32	.33	.62	.63
3	192.168.10.64	.65	.94	.95
4	192.168.10.96	.97	.126	.127
5	192.168.10.128	.129	.158	.159
6	192.168.10.160	.161	.190	.191
(7)	192.168.10.192	.193	.222	.223
(8)	192.168.10.224	.225	.254	.255

Subnets 7 and 8 are spare, available for future growth.

Example 9.5 — "I Need N Hosts Per Subnet"

Question: You need at least **50 hosts** in each subnet of `172.16.0.0/16`. What mask should you use?
Step 1 — Find host bits. We need $2^h - 2 \geq 50$.
$2^5 - 2 = 30$ (not enough), $2^6 - 2 = 62 \geq 50$ ✓ So $h =$ **6** host bits.
Step 2 — Prefix. $32 - 6 =$ **/26**, mask `255.255.255.192`
Step 3 — Verify. Each /26 gives 62 usable hosts — enough for 50 with room to grow.

The Mistake That Costs the Most Marks

Candidates round the wrong way.
For **subnets**, round the borrowed bits *up*: needing 6 subnets means 3 bits (8), not 2 bits (4).
For **hosts**, round the host bits *up* too: needing 50 hosts means 6 bits (62), not 5 bits (30).
And never forget the -2 when counting usable hosts.

9.6 VLSM — Variable Length Subnet Masking

With plain subnetting every subnet is the same size, which wastes addresses when departments differ in size. **VLSM** lets you use a *different* mask for each subnet.

The VLSM Golden Rule

Always allocate from the **largest requirement down to the smallest**. If you start with the small subnets, the blocks will not align and you will have to redo everything.

Example 9.6 — Full VLSM Design (Exam-Style)

A company has network `192.168.1.0/24` and these departments:
Admin 100 hosts • Sales 50 hosts • IT 25 hosts • HR 10 hosts • Router-to-router WAN link 2 hosts

Step 1 — Sort largest first and find the prefix each needs.

Dept	Needs	Host bits	Usable	Prefix
Admin	100	$2^7 - 2 = 126$	126	/25
Sales	50	$2^6 - 2 = 62$	62	/26
IT	25	$2^5 - 2 = 30$	30	/27
HR	10	$2^4 - 2 = 14$	14	/28
WAN	2	$2^2 - 2 = 2$	2	/30

Step 2 — Allocate blocks in order.

Dept	Network	Mask	First	Last	Broadcast
Admin	192.168.1.0/25	255.255.255.128	.1	.126	.127
Sales	192.168.1.128/26	255.255.255.192	.129	.190	.191
IT	192.168.1.192/27	255.255.255.224	.193	.222	.223
HR	192.168.1.224/28	255.255.255.240	.225	.238	.239
WAN	192.168.1.240/30	255.255.255.252	.241	.242	.243

Remaining free: `192.168.1.244` – `192.168.1.255` for future use.
Compare: with fixed-length subnetting you would have needed /25 for everyone and run out after two departments. VLSM fitted all five into the same /24.

Always Use /30 for Router-to-Router Links

A serial link between two routers needs exactly **2** usable addresses. A /30 provides exactly 2. Using anything larger wastes addresses. (Modern designs sometimes use /31 for point-to-point links, which gives 2 usable with no network/broadcast reservation.)

9.7 Supernetting and Route Aggregation

Supernetting is the opposite of subnetting: several small contiguous networks are combined into one larger block with a *shorter* prefix, so that routers advertise one route instead of many.

Example 9.7 — Supernetting

Combine these four Class C networks into one supernet:
`192.168.8.0/24`, `192.168.9.0/24`, `192.168.10.0/24`, `192.168.11.0/24`
Step 1 — Write the third octet in binary.

```
8  = 00001000
9  = 00001001
10 = 00001010
11 = 00001011
```

Step 2 — Find the common prefix. The first **6** bits (`000010`) are identical; only the last 2 bits vary.
Step 3 — Compute the new prefix. $8 + 8 + 6 = \mathbf{22}$
Answer: `192.168.8.0/22`, mask `255.255.252.0`
One routing table entry now replaces four.

9.8 NAT — Network Address Translation

NAT lets many devices with private IP addresses share one public IP address. The router rewrites the source address (and, in **PAT / NAT overload**, the source port too) on the way out, and reverses the translation on the way back.

Benefits: conserves scarce public IPv4 addresses, and hides the internal network structure (a security benefit).
Drawbacks: breaks true end-to-end connectivity, complicates peer-to-peer applications and VoIP, and adds processing overhead.

- **Static NAT** — one private IP permanently mapped to one public IP.
- **Dynamic NAT** — private IPs mapped to a pool of public IPs as needed.
- **PAT (Port Address Translation) / NAT overload** — many private IPs share a single public IP, distinguished by port numbers. This is what every home router does.

9.9 IPv6

IPv4's 32 bits gives about 4.3 billion addresses — long exhausted. **IPv6** uses **128 bits**, written as eight groups of four hexadecimal digits separated by colons.

$$2^{128} \approx 3.4 \times 10^{38} \text{ addresses}$$

Example: `2001:0db8:0000:0000:0000:ff00:0042:8329`
Shortened: `2001:db8::ff00:42:8329`

9.9.1 Shortening Rules

1. Drop *leading* zeros in each group: `0042` → `42`.
2. Replace one run of consecutive all-zero groups with `::` — but only **once** per address, otherwise it would be ambiguous.

Feature	IPv4	IPv6
Address size	32 bits	128 bits
Notation	Dotted decimal	Hexadecimal with colons
Header size	20–60 bytes, variable	40 bytes, fixed
Header fields	12	8
Checksum	Present in header	Removed (handled by other layers)
Fragmentation	By sender and routers	By sender only
Broadcast	Supported	**Not supported** — uses multicast
Configuration	Manual or DHCP	Stateless autoconfiguration (SLAAC)
Security	IPsec optional	IPsec built in
Address types	Unicast, multicast, broadcast	Unicast, multicast, anycast

IPv6 Facts Frequently Asked as One-Liners

Address length **128 bits** • Header **40 bytes fixed** • Loopback **::1** • Unspecified **::** • Link-local prefix **FE80::/10** • Multicast prefix **FF00::/8** • Global unicast **2000::/3** • **No broadcast, no ARP** (ARP is replaced by **Neighbour Discovery Protocol**).

9.9.2 Transition Mechanisms

- **Dual stack** — a device runs IPv4 and IPv6 at the same time.
- **Tunnelling** — IPv6 packets are wrapped inside IPv4 packets to cross an IPv4-only region.
- **Header translation (NAT64)** — headers are rewritten so an IPv6-only host can reach an IPv4-only host.

9.10 Special IPv4 Addresses — Quick Reference

Address / Range	Name	Purpose
0.0.0.0	This host / default route	Used at boot before an IP is assigned
127.0.0.0/8	Loopback	Testing the local TCP/IP stack
169.254.0.0/16	APIPA / link-local	Self-assigned when DHCP fails
255.255.255.255	Limited broadcast	All hosts on this local network
224.0.0.0/4	Multicast (Class D)	Group communication
240.0.0.0/4	Class E	Reserved for experiments

Final Exam Checklist for Any Subnetting Question

1. Write the mask in binary before doing anything else.
2. Block size $= 256 -$ interesting octet.
3. Network address = the block boundary at or below your host address.
4. Broadcast = next block boundary -1.
5. Usable hosts $= 2^h - 2$ — *never forget the minus two*.
6. For VLSM, always allocate largest first.
7. Show your binary working — most papers award marks for method, not just the final number.

CHAPTER 10

Routing

Asked In

What is routing? Explain different types of routing. — BPSC
What are static and dynamic routing? Give their relative advantages. — BPSC AE 2022
Define distance vector and link state routing protocols. — BPSC AE 2022
What is an autonomous system? — past written papers

10.1 What Routing Is

Routing is the process of selecting the best path for a packet to travel from source network to destination network. It happens at the **Network layer (Layer 3)** and is performed by **routers** using a **routing table**.

A routing table entry holds: destination network, subnet mask, next-hop address, outgoing interface, and a **metric** (cost).

10.2 Static vs Dynamic Routing

Static routing — routes entered manually by the administrator. They never change unless someone edits them.

Dynamic routing — routers exchange information with each other using routing protocols and build their tables automatically, adapting when links fail.

Point	Static Routing	Dynamic Routing
Configuration	Manual, by administrator	Automatic, by protocol
Adaptability	None — fails if a link drops	Reroutes automatically
CPU / memory use	Very low	Higher
Bandwidth overhead	None	Uses bandwidth for updates
Security	More secure (no advertisements)	Less secure
Scalability	Poor — unusable in large networks	Excellent
Best for	Small, stable networks; stub networks	Large, changing networks

Default route (`0.0.0.0/0`) — the "route of last resort", used when no specific entry matches. Very common on the router connecting an office to its ISP.

10.3 Distance Vector vs Link State

10.3.1 Distance Vector Routing

Each router tells its *direct neighbours* everything it knows about the whole network ("routing by rumour"). It uses the **Bellman-Ford** algorithm and the metric is usually **hop count**.

Updates are sent *periodically* (RIP: every 30 seconds) to neighbours only.

Problem: count-to-infinity. When a network goes down, bad information circulates and hop counts climb slowly. Remedies: **split horizon**, **poison reverse**, **hold-down timers**, and defining infinity as 16.

Protocols: RIP, RIPv2, IGRP, EIGRP (advanced/hybrid).

10.3.2 Link State Routing

Each router discovers its own directly connected links, then **floods** that information to *every* router in the area. Every router therefore builds an identical complete map of the topology and independently runs **Dijkstra's shortest path first** algorithm on it.

Updates are *event-triggered* — sent only when something changes.

Protocols: OSPF, IS-IS.

Point	Distance Vector	Link State
Algorithm	Bellman-Ford	Dijkstra (SPF)
Knowledge	Only what neighbours report	Full map of the topology
Shares with	Only direct neighbours	All routers (flooding)
Shares what	Its entire routing table	Only its own link states
Update timing	Periodic	Only when a change occurs
Metric	Hop count	Cost (bandwidth-based)
Convergence	Slow	Fast
CPU / memory	Low	High
Count-to-infinity	Yes, a real problem	No
Example	RIP, IGRP	OSPF, IS-IS

Easy Way to Remember

Distance vector = asking your neighbour for directions and trusting the answer. **Link state** = everyone has the full map, so each person works out the best route themselves. The map costs more memory but never gives you a stale rumour.

10.4 Autonomous Systems, IGP and EGP

An **Autonomous System (AS)** is a group of networks under a single administrative authority with a common routing policy — for example, the entire network of one ISP or one bank. Each AS has a globally unique **AS number**.

- **IGP (Interior Gateway Protocol)** — routes *within* one AS. Examples: RIP, OSPF, EIGRP, IS-IS.
- **EGP (Exterior Gateway Protocol)** — routes *between* different AS's. The only one used today is **BGP (Border Gateway Protocol)**, which is what holds the global Internet together. BGP is a **path vector** protocol.

Protocol	Type	Algorithm	Metric	Notes
RIP	IGP, DV	Bellman-Ford	Hop count	Max 15 hops; 16 = unreachable; simple, small networks
OSPF	IGP, LS	Dijkstra	Cost (bandwidth)	Open standard, uses areas, fast convergence
EIGRP	IGP, hybrid	DUAL	Composite	Cisco; fast, uses feasible successors
IS-IS	IGP, LS	Dijkstra	Cost	Used by large ISPs
BGP	EGP	Path vector	AS-path, policy	Routes the whole Internet

10.5 Other Routing Concepts

Unicast routing — one source to one destination.
Multicast routing — one source to a group (protocols: PIM, DVMRP).
Ad-hoc routing — for mobile networks (MANETs) with *no fixed infrastructure*, where nodes move and every node also acts as a router. Protocols: AODV, DSR, OLSR.

Benefit of ad-hoc routing: the network can form instantly wherever devices happen to be — no cabling, no base stations, no central administration. This makes it invaluable in disaster relief, military operations and rural areas where infrastructure does not exist or has been destroyed.

Routing vs Forwarding vs Switching

Routing = deciding *which path* a packet should take (building the table).
Forwarding = actually moving a packet from the input port to the correct output port using that table.
Switching at Layer 2 uses *MAC* addresses; routing at Layer 3 uses *IP* addresses.

CHAPTER 11

The Transport Layer

11.1 Purpose

The network layer delivers a packet to the right *host*. The transport layer delivers it to the right *process* on that host, using **port numbers**. This is called **process-to-process** delivery.

Socket = IP address + port number. It uniquely identifies one endpoint of a connection.

11.1.1 Port Number Ranges

- **Well-known ports:** 0–1023 (assigned to standard services)
- **Registered ports:** 1024–49151
- **Dynamic / ephemeral ports:** 49152–65535 (assigned temporarily to clients)

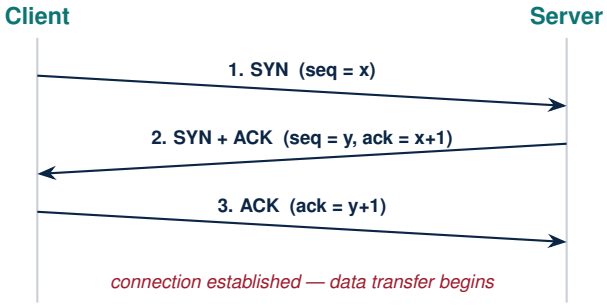
11.2 TCP vs UDP

Point	TCP	UDP
Full name	Transmission Control Protocol	User Datagram Protocol
Connection	Connection-oriented	Connectionless
Reliability	Reliable — guarantees delivery	Unreliable — best effort
Ordering	Guarantees order	No ordering guarantee
Acknowledgement	Yes	No
Retransmission	Yes, on loss	No
Flow control	Yes (sliding window)	No
Congestion control	Yes	No
Error checking	Checksum + recovery	Checksum only, no recovery
Header size	20–60 bytes	8 bytes (fixed)
Speed	Slower	Faster
Data unit	Segment	Datagram
Used by	HTTP, HTTPS, FTP, SMTP, Telnet, SSH	DNS, DHCP, TFTP, SNMP, VoIP, streaming, online games

Easy Way to Remember

TCP = registered post — slower, but you get a delivery confirmation and nothing is lost. **UDP** = dropping a postcard in the box — fast and cheap, but nobody confirms it arrived. Live video prefers UDP because a re-sent frame would arrive too late to be useful anyway.

11.3 The TCP Three-Way Handshake



1. **SYN** — the client sends a segment with the SYN flag set and its initial sequence number x.
2. **SYN + ACK** — the server replies with both flags, its own sequence number y, and acknowledges $x + 1$.
3. **ACK** — the client acknowledges $y + 1$. The connection is now open.

Connection termination uses a **four-way handshake**: FIN, ACK, FIN, ACK — because each direction must be closed separately.

11.4 Flow Control and Congestion Control

Flow control protects the *receiver* from being overwhelmed. TCP uses a **sliding window** whose size the receiver advertises in every segment.

Congestion control protects the *network* from being overwhelmed. TCP uses: **slow start** (window doubles each RTT), **congestion avoidance** (window grows linearly after a threshold), and **fast retransmit / fast recovery** on receiving three duplicate ACKs.

Common Mistake

Do not confuse the two. **Flow control = receiver's problem** (a slow PC). **Congestion control = network's problem** (a busy router). Exams often ask for the difference in one line.

CHAPTER 12

Application Layer Protocols

12.1 DNS — Domain Name System

Asked In

> *What is the function of DNS and DHCP?* — Rupali Bank ANE 2023

DNS translates human-readable **domain names** into **IP addresses** — it is the phone book of the Internet. It runs on **port 53**, usually over **UDP** (fast, small queries) and over TCP for large responses and zone transfers.

12.1.1 DNS Hierarchy

Root (.) → Top Level Domain (.com, .org, .bd) → Second level (google) → Subdomain (www)

12.1.2 Resolution Process

1. The client checks its local cache, then its `hosts` file.
2. It queries the configured **recursive resolver** (usually the ISP's DNS server).
3. The resolver asks a **root server**, which points to the **TLD server**.
4. The TLD server points to the **authoritative name server** for the domain.
5. The authoritative server returns the IP address, which is cached along the way for the period given by its **TTL**.

12.1.3 Common Record Types

Record	Purpose
A	Maps a name to an IPv4 address
AAAA	Maps a name to an IPv6 address
CNAME	Alias — points one name at another name
MX	Mail exchange — which server handles email for this domain
NS	Name server — which servers are authoritative for this zone
PTR	Reverse lookup — maps an IP address back to a name
TXT	Arbitrary text; used for SPF, DKIM and domain verification

12.2 DHCP — Dynamic Host Configuration Protocol

DHCP automatically assigns a device its IP address, subnet mask, default gateway and DNS server addresses. It runs on **UDP ports 67 (server) and 68 (client)**.

The DORA Process — Memorise This

Discover — the client broadcasts looking for any DHCP server.
Offer — servers reply offering an available IP address.
Request — the client broadcasts acceptance of one specific offer.
Acknowledge — the server confirms and records the **lease**.

Benefits: no manual configuration, no duplicate-address mistakes, addresses are recycled automatically when leases expire, and devices can move between networks freely.

Easy Way to Remember

DNS answers "*what is the address of this name?*"
DHCP answers "*what is my own address?*"
One looks outward, the other looks inward. This distinction is the entire answer to the Rupali Bank question above.

12.3 Well-Known Ports and Protocols

Protocol	Port	TCP/UDP	Function
FTP	20, 21	TCP	File transfer (20 = data, 21 = control)
SSH	22	TCP	Secure encrypted remote login
Telnet	23	TCP	Remote login — *unencrypted, insecure*
SMTP	25	TCP	Sending email
DNS	53	UDP/TCP	Name to IP resolution
DHCP	67, 68	UDP	Automatic IP configuration
TFTP	69	UDP	Trivial file transfer, no authentication
HTTP	80	TCP	Web browsing
POP3	110	TCP	Retrieve email (downloads and usually deletes)
IMAP	143	TCP	Retrieve email (keeps it on the server, syncs)
SNMP	161, 162	UDP	Network device management and monitoring
HTTPS	443	TCP	Secure web browsing with TLS
RDP	3389	TCP	Remote Desktop

Easy Way to Remember

Ports worth memorising as a block: **21 FTP, 22 SSH, 23 Telnet, 25 SMTP, 53 DNS, 80 HTTP, 110 POP3, 143 IMAP, 443 HTTPS**. These appear in almost every MCQ paper.

12.3.1 POP3 vs IMAP

Point	POP3	IMAP
Storage	Downloads to one device	Keeps mail on the server
Multi-device	Poor	Excellent — syncs everywhere
Server space	Frees it up	Uses it
Offline access	Full	Partial (cached)
Port	110 (995 secure)	143 (993 secure)

12.3.2 HTTP and HTTPS

HTTP is stateless and unencrypted. **HTTPS** is HTTP carried inside a **TLS/SSL** tunnel, providing *encryption* (nobody can read it), *authentication* (a certificate proves the server's identity) and *integrity* (tampering is detected).

CHAPTER 13

Network Security

13.1 The CIA Triad

- **Confidentiality** — only authorised people can read the data. Achieved by encryption.
- **Integrity** — data cannot be altered undetected. Achieved by hashing and digital signatures.
- **Availability** — the system is accessible when needed. Threatened by DoS attacks.

Two further goals are often added: **Authentication** (proving who you are) and **Non-repudiation** (you cannot later deny having sent it).

13.2 Cryptography

13.2.1 Symmetric Key Cryptography

The **same key** encrypts and decrypts. Fast and efficient for large volumes of data, but the key must somehow be shared securely in advance — which is the hard part.

For n users, $\frac{n(n-1)}{2}$ keys are needed.

Algorithms: DES (56-bit, now broken), 3DES, **AES** (128/192/256-bit, the current standard), Blowfish, RC4.

13.2.2 Asymmetric (Public Key) Cryptography

A **key pair**: a **public key** that everybody may know, and a **private key** kept secret. What one key encrypts, only the other can decrypt.

- For **confidentiality**: encrypt with the *receiver's public key*; only the receiver's private key can open it.
- For **a digital signature**: encrypt (sign) with the *sender's private key*; anyone can verify it with the sender's public key, proving it came from that sender.

For n users, only $2n$ keys are needed.

Algorithms: **RSA**, Diffie-Hellman (key exchange), ECC, DSA.

Point	Symmetric	Asymmetric
Keys	One shared secret key	Public + private key pair
Speed	Fast	Slow (100–1000×)
Key distribution	Difficult — the main weakness	Easy — publish the public key
Number of keys	$n(n-1)/2$	$2n$
Scalability	Poor	Good
Non-repudiation	Not provided	Provided via digital signature
Example	AES, DES, 3DES	RSA, ECC, Diffie-Hellman
Best for	Bulk data encryption	Key exchange and signatures

How Real Systems Combine Both — Hybrid Encryption

HTTPS uses *both*. Asymmetric cryptography (slow but solves key distribution) is used **once** at the start to safely agree a shared session key. Then symmetric cryptography (fast) encrypts all the actual traffic. You get the security of one and the speed of the other.

Easy Way to Remember

Think of a **public key** as an open padlock you hand out freely. Anyone can snap it shut on a box, but only you — holding the **private key** — can unlock it.

13.3 Hashing and Digital Signatures

A **hash function** (MD5, SHA-1, SHA-256) produces a fixed-length **digest** from any input. It is *one-way* — you cannot reverse it — and any change to the input completely changes the digest. Used for password storage and integrity checking.

A **digital signature** is the hash of a message, encrypted with the sender's private key. It provides authentication, integrity and non-repudiation all at once.

A **digital certificate** binds a public key to an identity and is issued by a trusted **Certificate Authority (CA)**. This is the basis of **PKI (Public Key Infrastructure)**.

13.4 Firewalls and VPN

A **firewall** filters traffic between networks according to rules.

- **Packet filtering** — inspects IP addresses, ports and protocol; stateless.
- **Stateful inspection** — tracks the state of connections; far more effective.
- **Application / proxy firewall** — understands the application protocol itself.
- **Next-generation firewall** — adds intrusion prevention and deep packet inspection.

A **VPN (Virtual Private Network)** creates an encrypted **tunnel** across the public Internet, so a remote branch or employee can safely reach the internal network. Protocols: **IPsec**, SSL/TLS VPN, L2TP, PPTP (obsolete and insecure).

DMZ (Demilitarised Zone) — an isolated network segment holding public-facing servers (web, mail), separated by firewalls from the internal network, so that a breach of a public server does not expose internal systems.

13.5 Common Attacks

Attack	Description
DoS / DDoS	Flooding a service with traffic until it cannot serve legitimate users. DDoS uses many compromised machines (a botnet).
Man-in-the-middle	The attacker secretly relays and possibly alters traffic between two parties who think they are talking directly.
Phishing	Fraudulent messages that trick a user into revealing credentials.
IP spoofing	Forging the source IP address of packets.
ARP poisoning	Sending false ARP replies so traffic is diverted through the attacker.
SQL injection	Inserting malicious SQL into an input field to manipulate the database.
Sniffing	Passively capturing traffic on a shared medium.
Brute force	Trying every possible password until one works.
Ransomware	Encrypting the victim's data and demanding payment for the key.

Point	Active Attack	Passive Attack
Nature	Modifies or disrupts data	Only observes and copies
Detection	Easier to detect	Very hard to detect
Prevention	Hard to prevent	Prevented by encryption
Examples	DoS, MITM, spoofing, replay	Sniffing, traffic analysis

CHAPTER 14

Wireless and Mobile Networks

14.1 IEEE 802 Standards

Standard	Description
802.3	Ethernet (CSMA/CD)
802.4	Token Bus
802.5	Token Ring
802.11	Wireless LAN (Wi-Fi)
802.15	Wireless PAN — 802.15.1 Bluetooth, 802.15.4 Zigbee
802.16	WiMAX (broadband wireless MAN)

14.2 Wi-Fi (802.11) Generations

Standard	Wi-Fi Name	Frequency	Max Speed	Notes
802.11a	—	5 GHz	54 Mbps	Shorter range
802.11b	—	2.4 GHz	11 Mbps	Longer range, crowded band
802.11g	—	2.4 GHz	54 Mbps	Backward compatible with b
802.11n	Wi-Fi 4	2.4 & 5 GHz	600 Mbps	Introduced MIMO
802.11ac	Wi-Fi 5	5 GHz	6.9 Gbps	MU-MIMO
802.11ax	Wi-Fi 6	2.4 & 5 GHz	9.6 Gbps	OFDMA, better in crowds
802.11be	Wi-Fi 7	2.4/5/6 GHz	46 Gbps	Multi-link operation

14.2.1 Wireless Security

WEP (broken, never use) → **WPA** (TKIP) → **WPA2** (AES-CCMP, the long-time standard) → **WPA3** (SAE handshake, current best).

14.2.2 Wireless Network Modes

Infrastructure mode — devices connect through an **access point**.
Ad-hoc mode — devices connect directly to each other with no access point.

SSID is the network name. **BSS** is one access point and its clients; **ESS** is several BSSs joined by a distribution system.

14.3 Bluetooth

Short-range (about 10 m for Class 2), 2.4 GHz, uses **FHSS**. Devices form a **piconet**: one **master** and up to **7 active slaves**. Interconnected piconets form a **scatternet**.

14.4 Mobile Generations

Asked In

Describe the protocols for second-generation (2G) digital cellular networks used by mobile operators. — past written papers

Gen	Era	Technology	Speed	Key Feature
1G	1980s	AMPS, analog	2.4 kbps	Analog voice only
2G	1991	**GSM**, CDMA (IS-95), TDMA	64 kbps	First *digital*; SMS; encryption; SIM cards
2.5G	—	GPRS	114 kbps	Always-on packet data
2.75G	—	EDGE	384 kbps	Faster data on GSM
3G	2001	UMTS, WCDMA, HSPA	2 Mbps+	Video calling, mobile Internet
4G	2009	LTE, LTE-Advanced	100 Mbps – 1 Gbps	All-IP, HD streaming, VoLTE
5G	2019	NR (New Radio)	10–20 Gbps	Very low latency, massive IoT, network slicing

2G in Detail — The Answer This Question Wants

2G was the move from **analog to digital**. The dominant standard is **GSM (Global System for Mobile Communications)**, which uses a combination of **TDMA** and **FDMA** in the 900/1800 MHz bands.
Its major contributions: digital voice encoding, **SMS** text messaging, encrypted radio links, and the **SIM card**, which separated the subscriber identity from the handset for the first time.
The competing 2G family was **CDMA (IS-95)**, which used spread spectrum so that all users shared the whole band, each identified by a unique code.
Data extensions: **GPRS** (2.5G, packet switched) and **EDGE** (2.75G).

Point	GSM	CDMA
Access method	TDMA + FDMA	Code division (spread spectrum)
Subscriber identity	SIM card — easily swapped	Tied to the handset (historically)
Global coverage	Very widespread	Fewer countries
Capacity	Lower	Higher

CHAPTER 15

Formula Sheet and Rapid Revision

15.1 Every Formula in One Place

Quantity	Formula	Units
Frequency	$f = 1/T$	Hz
Period	$T = 1/f$	seconds
Bit rate	Baud rate $\times \log_2 L$	bps
Baud rate	Bit rate $/ \log_2 L$	baud
Nyquist (noiseless)	$C = 2B \log_2 L$	bps
Shannon (noisy)	$C = B \log_2(1 + \text{SNR})$	bps
Attenuation / gain	$\text{dB} = 10 \log_{10}(P_2/P_1)$	dB
SNR in dB	$10 \log_{10}(\text{SNR})$	dB
Sampling rate (Nyquist)	$f_s \geq 2f_{\max}$	samples/s
PCM bit rate	$f_s \times$ bits per sample	bps
Bits per sample	$\log_2(\text{quantization levels})$	bits
Propagation time	Distance / propagation speed	seconds
Transmission time	Message size / bandwidth	seconds
Mesh links	$n(n-1)/2$	links
Mesh ports per device	$n - 1$	ports
Number of subnets	2^s (s = borrowed bits)	—
Usable hosts per subnet	$2^h - 2$ (h = host bits)	—
Block size	$256-$ interesting mask octet	—
Hamming redundant bits	$2^r \geq m + r + 1$	bits
Symmetric keys needed	$n(n-1)/2$	keys
Asymmetric keys needed	$2n$	keys
Go-Back-N window	$2^m - 1$	frames
Selective Repeat window	2^{m-1}	frames

15.2 Subnet Mask Reference Table — Memorise This

CIDR	Subnet Mask	Block	Hosts	CIDR	Subnet Mask
/8	255.0.0.0	—	16,777,214	/24	255.255.255.0
/16	255.255.0.0	—	65,534	/25	255.255.255.128
/20	255.255.240.0	—	4,094	/26	255.255.255.192
/21	255.255.248.0	—	2,046	/27	255.255.255.224
/22	255.255.252.0	—	1,022	/28	255.255.255.240
/23	255.255.254.0	—	510	/29	255.255.255.248
/24	255.255.255.0	256	254	/30	255.255.255.252

Last-octet quick table: /25 → 128 (126 hosts) /26 → 192 (62) /27 → 224 (30) /28 → 240 (14) /29 → 248 (6) /30 → 252 (2)

15.3 One-Line Answers to the Most Repeated Questions

Question	One-Line Answer
Difference: half vs full duplex	Half = both ways but one at a time; full = both ways simultaneously
Why packet switching for data	Computer traffic is bursty; shared links are not wasted during idle periods, and packets reroute around failures
Switch vs router	Switch = Layer 2, MAC, breaks collision domains; router = Layer 3, IP, breaks broadcast domains
Why CSMA/CA in wireless	A radio cannot hear a collision while transmitting, so collisions must be avoided rather than detected
Classful → classless	To stop wasting addresses, slow IPv4 exhaustion, and allow route aggregation
DNS vs DHCP	DNS: name → IP. DHCP: assigns your own IP automatically (DORA)
TCP vs UDP	TCP reliable, ordered, connection-oriented; UDP fast, connectionless, no guarantee
Distance vector vs link state	DV: tells neighbours everything, periodic, Bellman-Ford. LS: tells everyone about itself, event-driven, Dijkstra
Symmetric vs asymmetric	Same key, fast, hard to distribute vs key pair, slow, easy to distribute
Bandwidth vs throughput	Bandwidth = theoretical maximum; throughput = actual measured rate
Bit rate vs baud rate	Bits per second vs signal elements per second; equal only when 1 bit per symbol
ARP vs RARP	ARP: IP → MAC. RARP: MAC → IP
Radio vs microwave	Radio omnidirectional, penetrates walls; microwave unidirectional, line of sight
Baseband vs broadband	Baseband: one digital signal, whole cable, TDM. Broadband: many analog channels, FDM

15.4 Numbers You Must Not Forget

- IPv4 = 32 bits IPv6 = 128 bits
- MAC address = 48 bits (12 hex digits)
- IPv4 header = 20–60 bytes; IPv6 = 40 bytes fixed
- TCP header = 20–60 bytes; UDP = 8 bytes
- Ethernet frame = 64 to 1518 bytes; MTU = 1500 bytes
- Class A: 1–126 B: 128–191 C: 192–223
- Loopback = 127.0.0.1 APIPA = 169.254.x.x
- Pure ALOHA = 18.4% Slotted ALOHA = 36.8%
- RIP max hop count = 15 (16 = unreachable)
- Bluetooth piconet = 1 master + 7 active slaves
- Voice channel (DS-0) = 64 kbps
- UTP maximum run = 100 metres
- GEO satellite altitude = 35,786 km
- OSI layers = 7 TCP/IP layers = 4
- Well-known ports = 0–1023

CHAPTER 16

MCQ Practice Set

Attempt all 60 questions in 45 minutes. Answers are at the end of the chapter.

1. How many layers are in the OSI model?
   (a) 4 (b) 5 (c) 6 (d) 7
2. Which layer is responsible for routing?
   (a) Data Link (b) Network (c) Transport (d) Session
3. The data unit at the Data Link layer is a
   (a) bit (b) frame (c) packet (d) segment
4. A router operates at OSI layer
   (a) 1 (b) 2 (c) 3 (d) 7
5. Which device breaks up broadcast domains?
   (a) hub (b) switch (c) router (d) repeater
6. How many bits are in an IPv4 address?
   (a) 16 (b) 32 (c) 64 (d) 128
7. How many bits are in an IPv6 address?
   (a) 32 (b) 64 (c) 128 (d) 256
8. The default subnet mask of Class B is
   (a) 255.0.0.0 (b) 255.255.0.0 (c) 255.255.255.0 (d) 255.255.255.255
9. Which is a private IP address?
   (a) 172.32.5.1 (b) 192.169.1.1 (c) 10.5.4.3 (d) 11.0.0.1
10. The loopback address is
    (a) 127.0.0.1 (b) 0.0.0.0 (c) 255.255.255.255 (d) 192.168.0.1
11. How many usable hosts in a /26 network?
    (a) 30 (b) 62 (c) 64 (d) 126
12. The subnet mask for /28 is
    (a) 255.255.255.192 (b) 255.255.255.224 (c) 255.255.255.240 (d) 255.255.255.248
13. Which protocol resolves IP to MAC?
    (a) RARP (b) ARP (c) DHCP (d) DNS
14. DNS uses port
    (a) 25 (b) 53 (c) 80 (d) 110
15. HTTPS uses port
    (a) 80 (b) 443 (c) 8080 (d) 21
16. Which is connectionless?
    (a) TCP (b) UDP (c) FTP (d) SMTP
17. The TCP handshake has how many steps?
    (a) 2 (b) 3 (c) 4 (d) 5
18. UDP header size is
    (a) 8 bytes (b) 20 bytes (c) 40 bytes (d) 60 bytes
19. Which protocol is used to send email?
    (a) POP3 (b) IMAP (c) SMTP (d) SNMP
20. DHCP process is known as
    (a) DORA (b) ARQ (c) SPF (d) DUAL
21. Which topology needs $n(n-1)/2$ links?
    (a) star (b) bus (c) ring (d) mesh
22. In star topology the single point of failure is
    (a) cable (b) NIC (c) central hub (d) terminator
23. Wired Ethernet uses
    (a) CSMA/CA (b) CSMA/CD (c) token passing (d) polling
24. Maximum efficiency of pure ALOHA is
    (a) 18.4% (b) 36.8% (c) 50% (d) 100%
25. 10Base5 has a maximum segment length of
    (a) 100 m (b) 185 m (c) 500 m (d) 2000 m
26. Manchester encoding always has a transition
    (a) at the start (b) in the middle (c) at the end (d) never
27. Which line code has no DC component?
    (a) unipolar NRZ (b) AMI (c) NRZ-L (d) NRZ-I
28. PCM stands for
    (a) Pulse Carrier Mod. (b) Pulse Code Mod. (c) Phase Code Mod. (d) Packet Control Mode
29. The first step of PCM is
    (a) quantization (b) encoding (c) sampling (d) modulation
30. Nyquist sampling rate must be at least
    (a) $f_{\max}$ (b) $2f_{\max}$ (c) $4f_{\max}$ (d) $f_{\max}/2$
31. A telephone voice channel is
    (a) 32 kbps (b) 64 kbps (c) 128 kbps (d) 256 kbps
32. ASK changes the signal's
    (a) amplitude (b) frequency (c) phase (d) all three
33. QAM changes
    (a) amplitude only (b) phase only (c) amplitude and phase (d) frequency only
34. FDM divides the link by
    (a) time (b) frequency (c) code (d) wavelength
35. WDM is used with
    (a) twisted pair (b) coaxial (c) fibre optic (d) radio
36. Which medium is immune to EMI?
    (a) UTP (b) STP (c) coaxial (d) fibre optic
37. Cat6 supports a bandwidth of
    (a) 100 MHz (b) 250 MHz (c) 500 MHz (d) 600 MHz
38. The connector used with UTP is
    (a) BNC (b) RJ-11 (c) RJ-45 (d) SC
39. Fibre works on the principle of
    (a) refraction (b) total internal reflection (c) diffraction (d) absorption
40. Microwave transmission requires
    (a) omnidirectional antenna (b) line of sight (c) no antenna (d) copper wire
41. Circuit switching is used in
    (a) Internet (b) telephone network (c) email (d) LAN
42. Which is a store-and-forward technique?
    (a) circuit switching (b) message switching (c) simplex (d) FDM
43. CRC uses which operation?
    (a) AND (b) OR (c) XOR (d) NOT
44. Simple parity cannot detect
    (a) 1-bit error (b) 3-bit error (c) 2-bit error (d) 5-bit

error

45. Hamming code can correct
(a) no error (b) single-bit error (c) burst error (d) all errors
46. Which retransmits only the damaged frame?
(a) stop-and-wait (b) Go-Back-N (c) Selective Repeat (d) ALOHA
47. RIP uses which metric?
(a) bandwidth (b) hop count (c) delay (d) load
48. OSPF uses which algorithm?
(a) Bellman-Ford (b) Dijkstra (c) DUAL (d) RSA
49. RIP's maximum hop count is
(a) 10 (b) 15 (c) 16 (d) 32
50. BGP is which type of protocol?
(a) IGP (b) EGP (c) link state (d) distance vector
51. Which is an asymmetric algorithm?
(a) AES (b) DES (c) RSA (d) Blowfish
52. A digital signature uses the sender's
(a) public key (b) private key (c) shared key (d) session key
53. How many keys for n users in symmetric crypto?
(a) n (b) $2n$ (c) $n(n-1)/2$ (d) n^2
54. Which attack is passive?
(a) DoS (b) sniffing (c) spoofing (d) MITM
55. Wi-Fi is defined by
(a) 802.3 (b) 802.5 (c) 802.11 (d) 802.16
56. Bluetooth piconet supports how many active slaves?
(a) 3 (b) 5 (c) 7 (d) 10
57. Which generation introduced SMS?
(a) 1G (b) 2G (c) 3G (d) 4G
58. GSM uses which access method?
(a) CDMA (b) TDMA + FDMA (c) OFDMA (d) ALOHA
59. Which is NOT in IPv6?
(a) unicast (b) multicast (c) anycast (d) broadcast
60. Bandwidth of a channel from 200 Hz to 3400 Hz is
(a) 3400 Hz (b) 3200 Hz (c) 3600 Hz (d) 200 Hz

Answer Key

1-d 2-b 3-b 4-c 5-c 6-b 7-c 8-b 9-c 10-a 11-b 12-c 13-b 14-b 15-b 16-b 17-b 18-a 19-c 20-a 21-d 22-c 23-b 24-a 25-c 26-b 27-b 28-b 29-c 30-b 31-b 32-a 33-c 34-b 35-c 36-d 37-b 38-c 39-b 40-b 41-b 42-b 43-c 44-c 45-b 46-c 47-b 48-b 49-b 50-b 51-c 52-b 53-c 54-b 55-c 56-c 57-b 58-b 59-d 60-b

CHAPTER 17

Written Question Bank

These are the question forms that actually appear. Practise writing each answer in the space a real answer sheet gives you — usually half a page for a 5-mark question.

17.1 Definition and Short-Note Questions

1. What is data communication? Define simplex, half-duplex and full-duplex.

Cover: definition, the four characteristics, then one short paragraph and one example each for the three modes. Draw the three-arrow diagram — it earns marks on its own.

2. Describe the data communication system with a necessary diagram.

Draw the five-component diagram (message, sender, receiver, medium, protocol), then explain each in one line. This is a guaranteed-mark question if you draw the figure.

3. Distinguish between synchronous and asynchronous transmission.

Answer in a table with at least six rows: unit of transfer, start/stop bits, gaps, timing, speed, overhead, example.

4. Write short notes on 10Base2, 10Base5, 10BaseT, 10BaseF, 100BaseT, 1000BaseTX.

Give speed, medium, maximum length and topology for each. A table is faster to write and easier to mark than prose.

5. Explain the terms domain, bandwidth, broadcast and multicast.

Define collision domain and broadcast domain, both meanings of bandwidth (Hz and bps), broadcast (to all, FF:FF:FF:FF:FF:FF) and multicast (to a group, Class D).

17.2 Comparison Questions — Always Answer in a Table

6. Circuit switching vs packet switching. Which is better and why?

Ten-row table, then a concluding paragraph: neither is universally better — circuit switching suits continuous real-time voice, packet switching suits bursty digital data, which is why the Internet uses it.

7. Mesh topology vs star topology.

Compare cabling cost ($n(n-1)/2$ vs n), reliability, single point of failure, fault isolation, and scalability.

8. Client-server vs peer-to-peer network.

Compare roles, storage, security, cost, scalability, backup, failure behaviour.

9. Twisted pair vs coaxial vs fibre optic.

Three-column table: signal type, bandwidth, data rate, attenuation, noise immunity, distance, cost, security.

10. Static routing vs dynamic routing; distance vector vs link state.

Two separate tables. For DV vs LS remember: *who* they talk to, *what* they send, *when* they send it, and *which algorithm* they run.

11. TCP vs UDP.

Twelve-row table plus one line on when to choose each.

12. Collision domain vs broadcast domain.

Define both, then give the hub/switch/router table, then the one-line rule: switches break collision domains, routers break broadcast domains.

17.3 Numerical Questions

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13. Convert a decimal IP address into binary.

Use the 128-64-32-16-8-4-2-1 subtraction method. Show each octet's working.

14. Given an IP and subnet mask, find network address, broadcast address and valid host range.

Follow the five-step method. Show the binary AND operation — method marks are usually worth more than the final answer.

15. Design subnets for departments of given sizes (VLSM).

Sort largest first, compute the prefix each needs from $2^h - 2$, allocate contiguously, then present a table with network, mask, first host, last host and broadcast for each department.

16. Calculate the maximum data rate of a channel.

Identify whether it is Nyquist (levels given) or Shannon (SNR given). If both are given, compute both and take the smaller.

17. Find the CRC for given data and generator.

Append $n - 1$ zeros, divide with XOR, show every step of the long division, state the remainder as the CRC, then write the transmitted frame.

18. Construct a Hamming code / locate the erroneous bit.

Find r from $2^r \geq m + r + 1$, place parity bits at powers of 2, compute each with even parity, and for error location build the syndrome from the failed checks.

17.4 Descriptive Questions

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19. Describe the OSI model with the function of each layer.

Draw the seven-layer stack. For each layer give: name, one-line function, data unit, and one device or protocol. Mention encapsulation at the end.

20. What is routing? Explain its types. What is the benefit of ad-hoc routing?

Define routing and the routing table, cover static vs dynamic, then unicast/multicast/ad-hoc. For ad-hoc, stress: no fixed infrastructure, every node also routes, instant deployment in disaster or military situations.

21. What is modulation? Why is it necessary? Describe PCM.

Definition, then the four reasons (antenna size, distance, multiplexing, noise), then the three PCM steps with the 64 kbps voice-channel calculation as a worked example.

22. What is a subnet? What benefits will you get from using subnets in this office?

Definition, then the benefits: reduced broadcast traffic, improved performance, better security through isolation, simpler management and troubleshooting, and efficient address use. Follow with a worked subnet design if the question gives department sizes.

23. What is the function of DNS and DHCP?

DNS: hierarchical name-to-IP resolution, port 53, record types, resolution steps. DHCP: automatic configuration, DORA, leases, port 67/68. Finish with the one-line contrast.

24. Describe the protocols used in 2G digital cellular networks.

GSM (TDMA + FDMA, SIM card, SMS, encryption), CDMA/IS-95 (spread spectrum), then the data extensions GPRS (2.5G) and EDGE (2.75G).

Written Exam Technique

1. Start every answer with a one-sentence definition, even for comparison questions.
2. Use tables for every "differentiate" or "distinguish" question — they are faster to write and easier for the examiner to award marks from.
3. Draw the diagram whenever one exists. Diagrams carry independent marks.
4. For numericals, show every step. Method marks survive an arithmetic slip.
5. Give a Bangladeshi or everyday example where you can — it shows understanding rather than memorisation.
6. Manage time: roughly 1.5 minutes per mark. Do not spend fifteen minutes perfecting a five-mark answer.

Final Word

This subject rewards *practice*, not re-reading. Work through every numerical in this book with pen and paper at least twice, and make sure you can produce the subnet mask table from memory. Those two things alone account for the largest block of marks available in the paper.